

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: DispM-Rev-Cg.sil etc.

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For: Spruce

32 total sheets = 31 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

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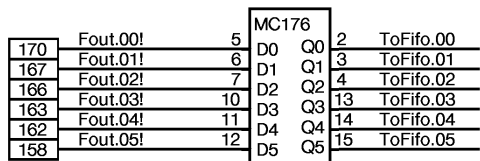
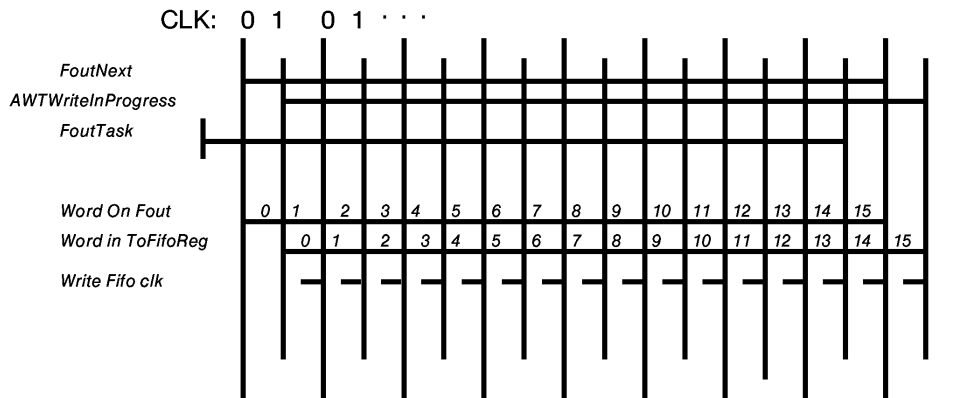
Not impl.: brightness, hue, saturation, show-object, show-dots-opaque, dots from files.

DORADO SCHEMATICS

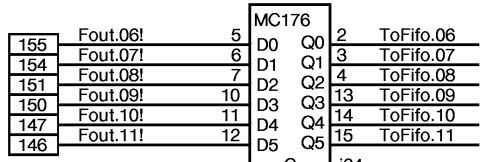
Display M

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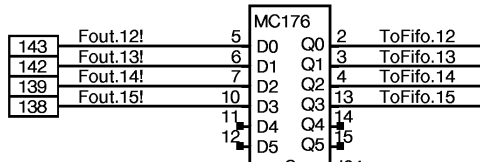
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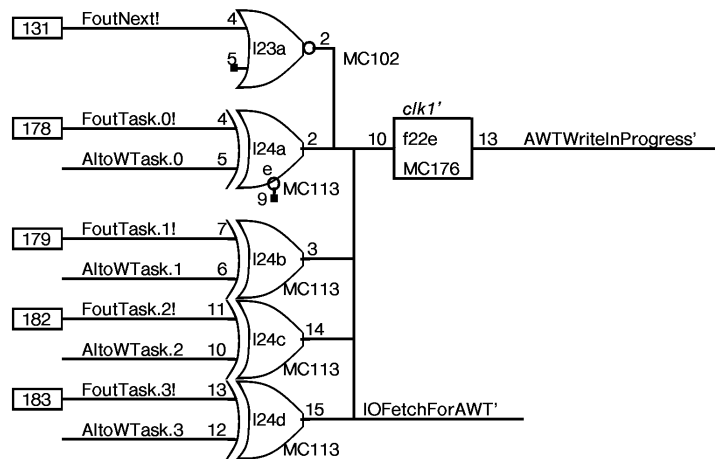
clk2'Da 9



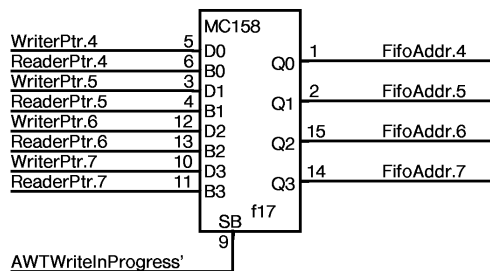
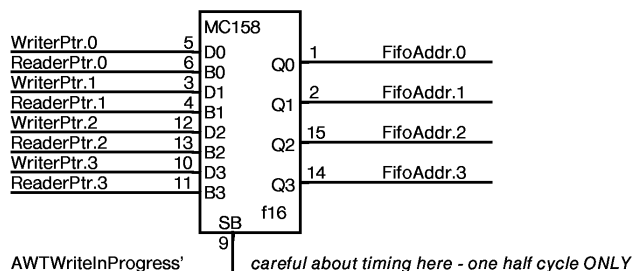
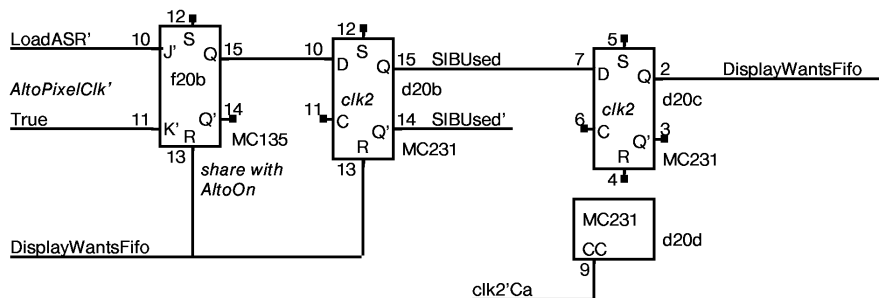
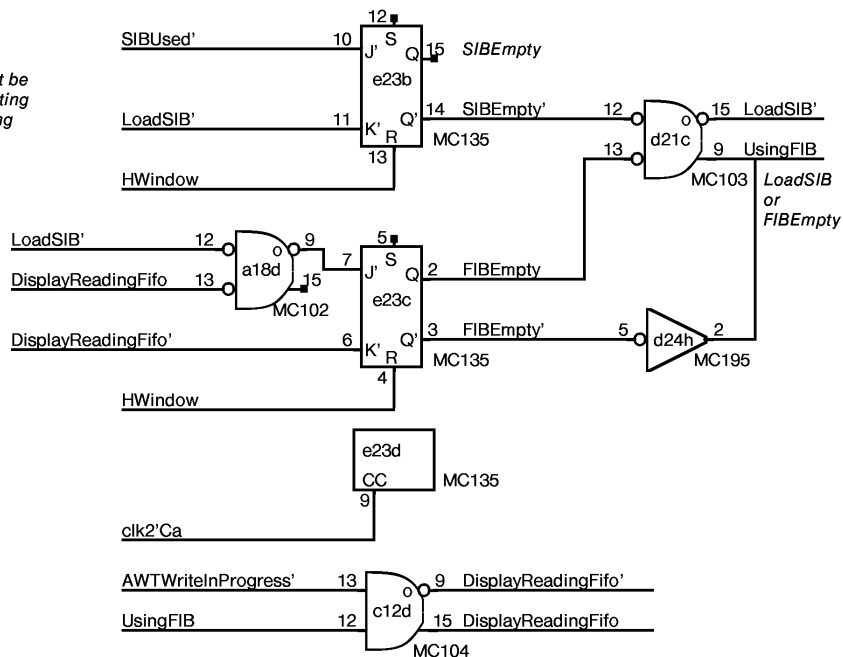
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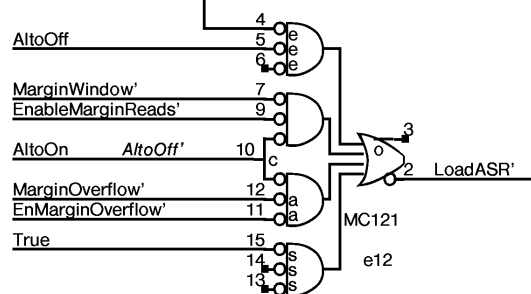
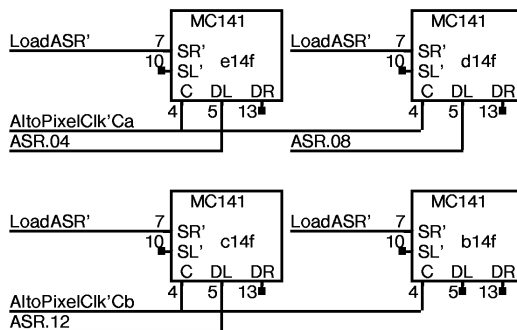
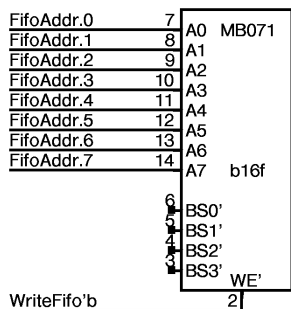
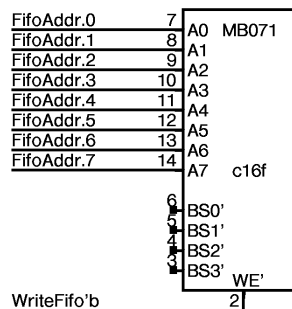
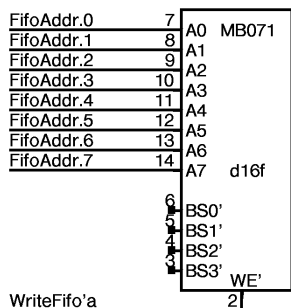
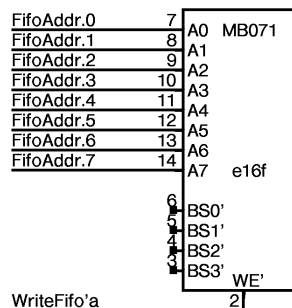
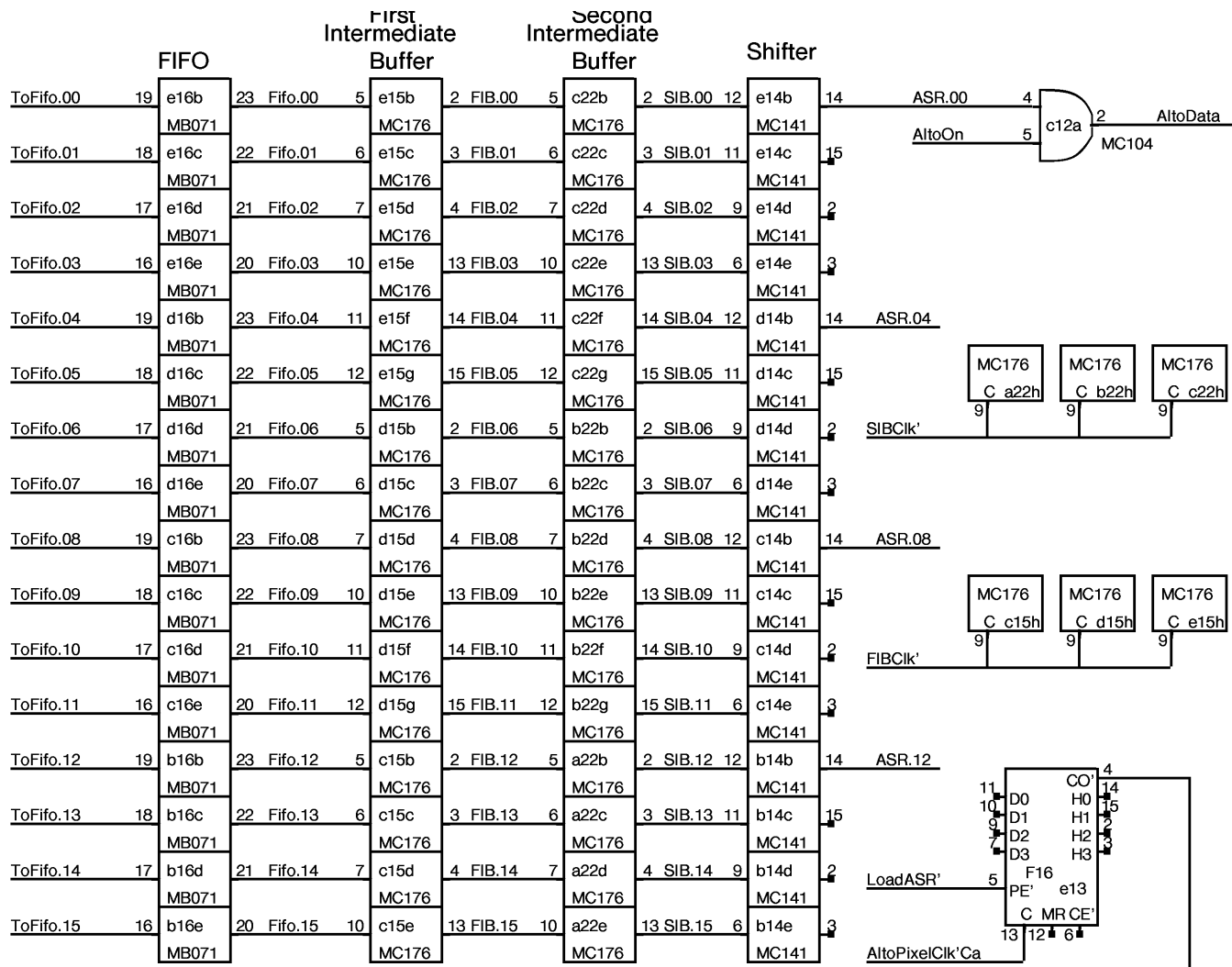


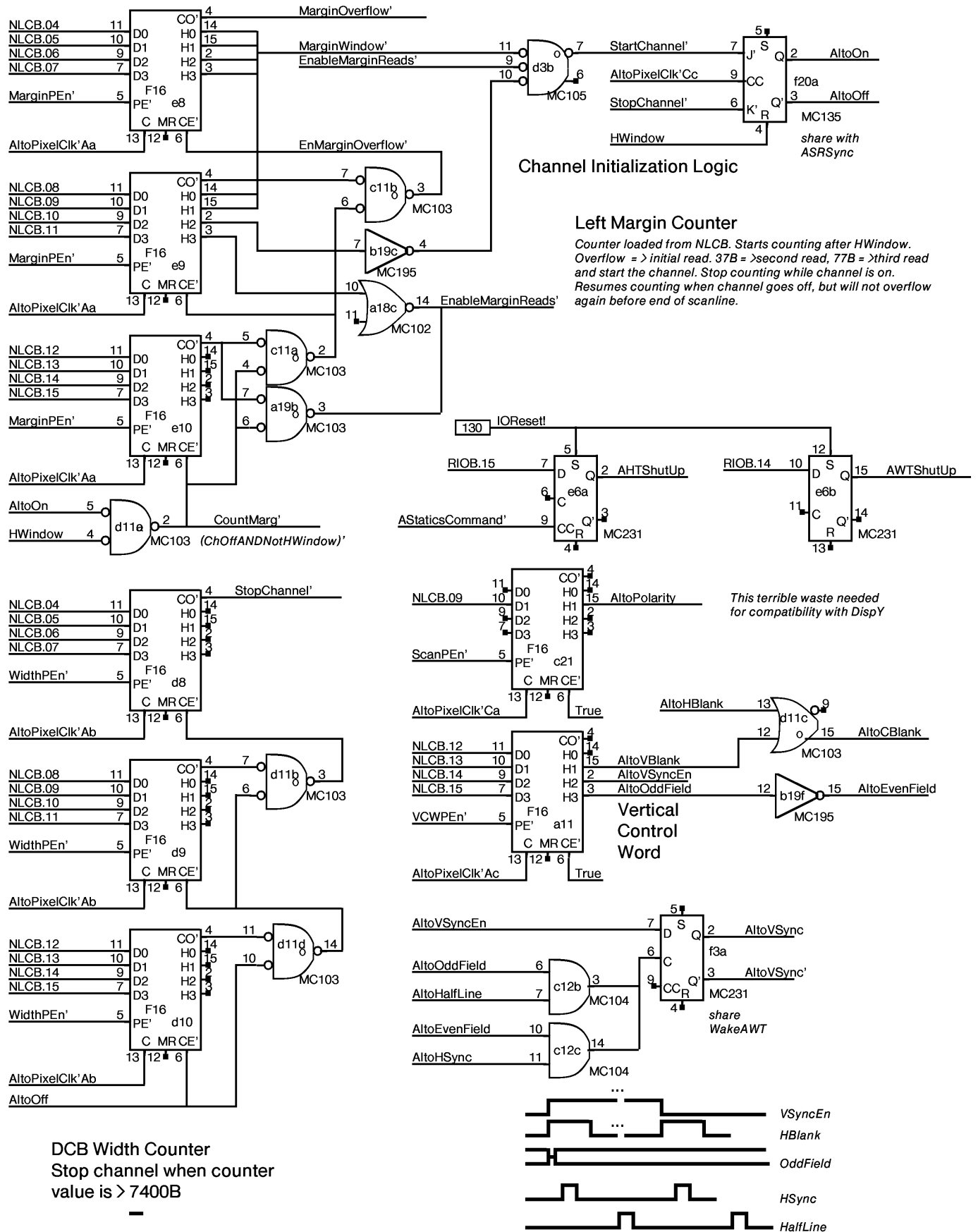
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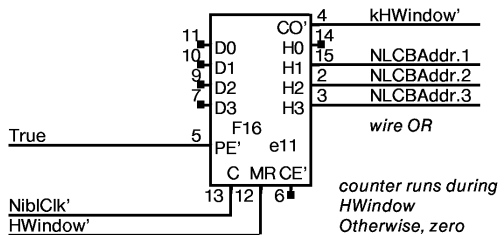
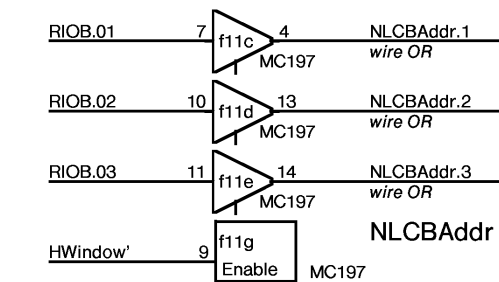


The figure shows two identical 32-bit write ports. Each port has four data inputs (D0-D3), a 16-bit address input (F16), a write enable input (PE'), a 3-bit burst length input (f18 or f19), a cache control input (C), and a memory refresh control input (MRCE'). The top port has four output pointers (H0-H3) and four corresponding WriterPtr outputs (WriterPtr.0-3). The bottom port has four output pointers (H0-H3) and four corresponding WriterPtr outputs (WriterPtr.4-7). Both ports share common control inputs: True, IncrementWPtr'a, and AltoVSync.

[illegible]

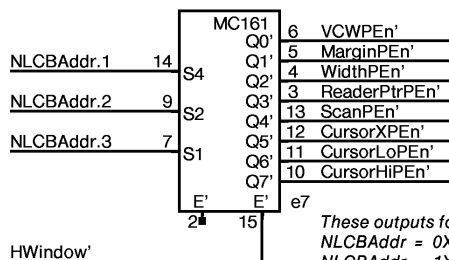






Note: rate is Pixel/4, unlike DispY which is Pixel/2. Microcode compensates.

CLCB load enables

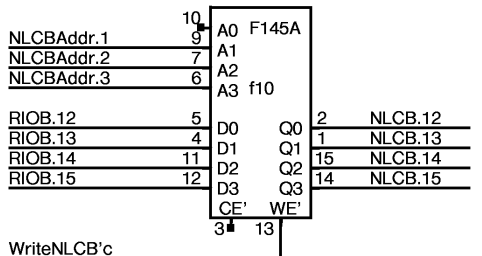
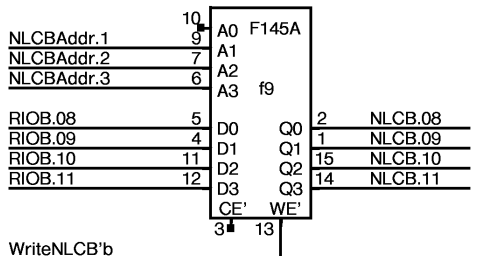
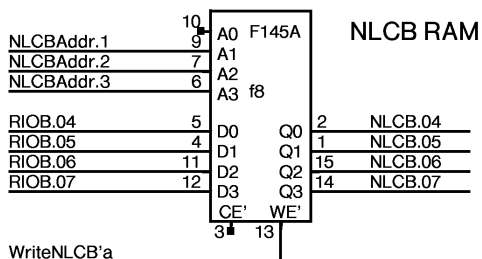


CursorXPE'n must occur before CursorLo/HiPE'n

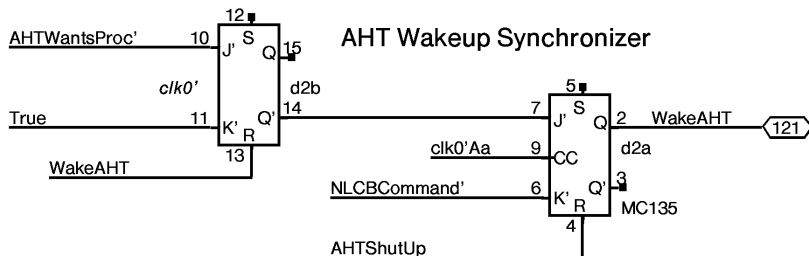
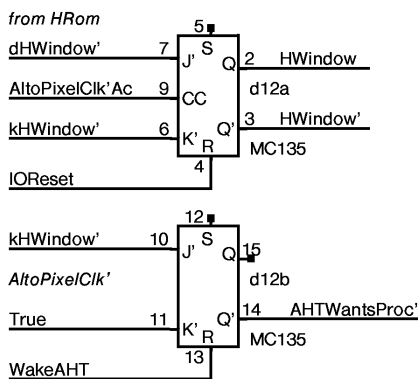
These outputs for either NLCBAddr = 0XXX or NLCBAddr = 1XXX Needed for compatibility with DispY

NOTE:

There are only 8 entries in NLCB.
The MSB (NLCBAddr.0) is ignored.
Outputs to 0XXX or 1XXX write words 0XXX of NLCB. During HWindow, CLCB entries are redundantly written twice, first when counter goes 00-07 and again when counter goes 10-17.

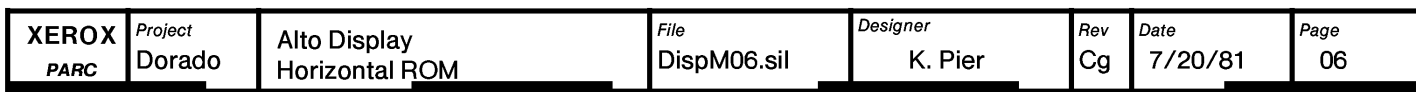


HWindow



For convenience, use NLCB command to kill wakeup. Assumes AHT will always do some NLCB command whenever it is awakened. Default would be to redundantly load NLCB[0] with same data.

requires minimum 4 instruction loop



NEXT Word Control Block Flag (NextWCBFlag)
Current Word Control Block Flag (CurrentWCBFlag)

A Word Control Block is a pair of values called
Address and MunchCount, for either the CURRENT
or the NEXT scanline.

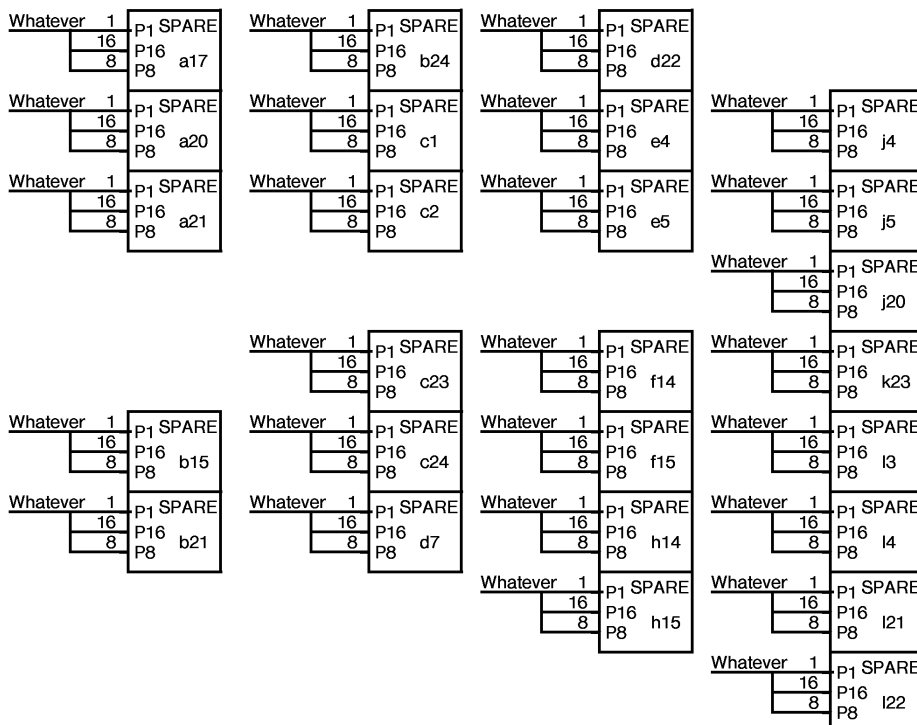
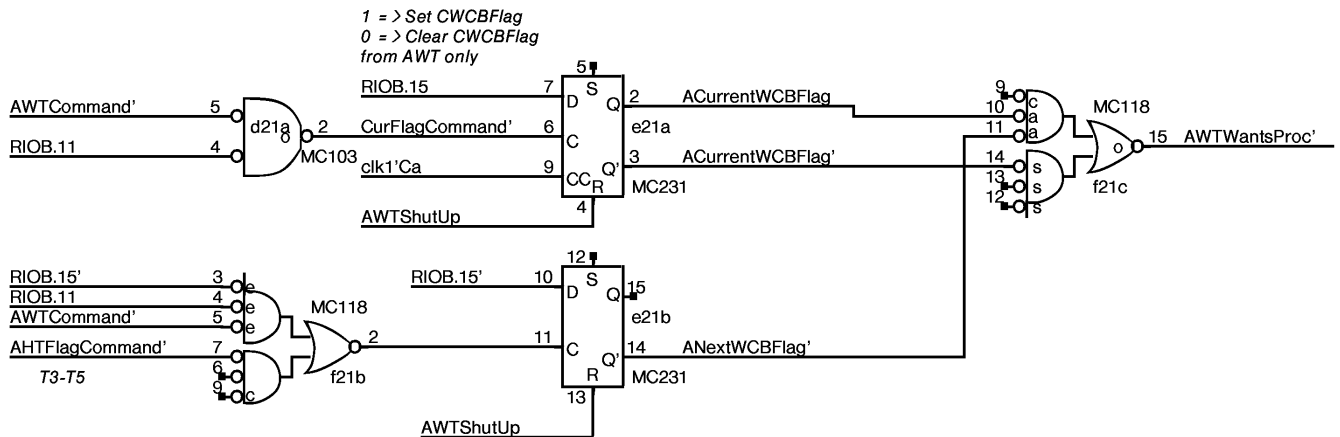
For AWT Commands:		For AHT commands:	
RIOB	Means	RIOB	Means
= 1c	Set CWCBFlag and Clear NWCBFlag	= 0c	Set ANextWCBFlag
= 0c	Clear CWCBFlag		

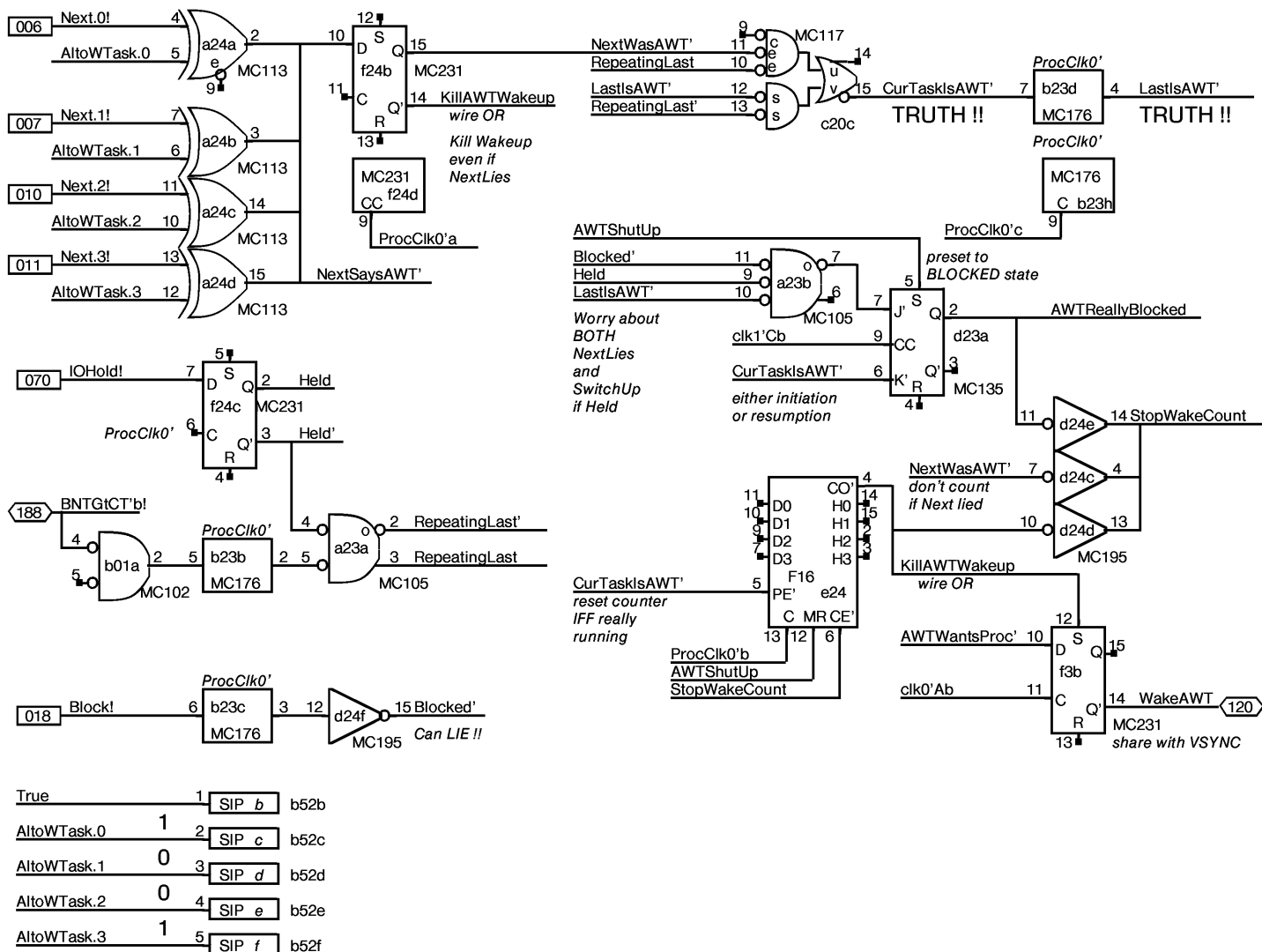
WakeUp conditions:

WakeAHT: at end of every HWindow

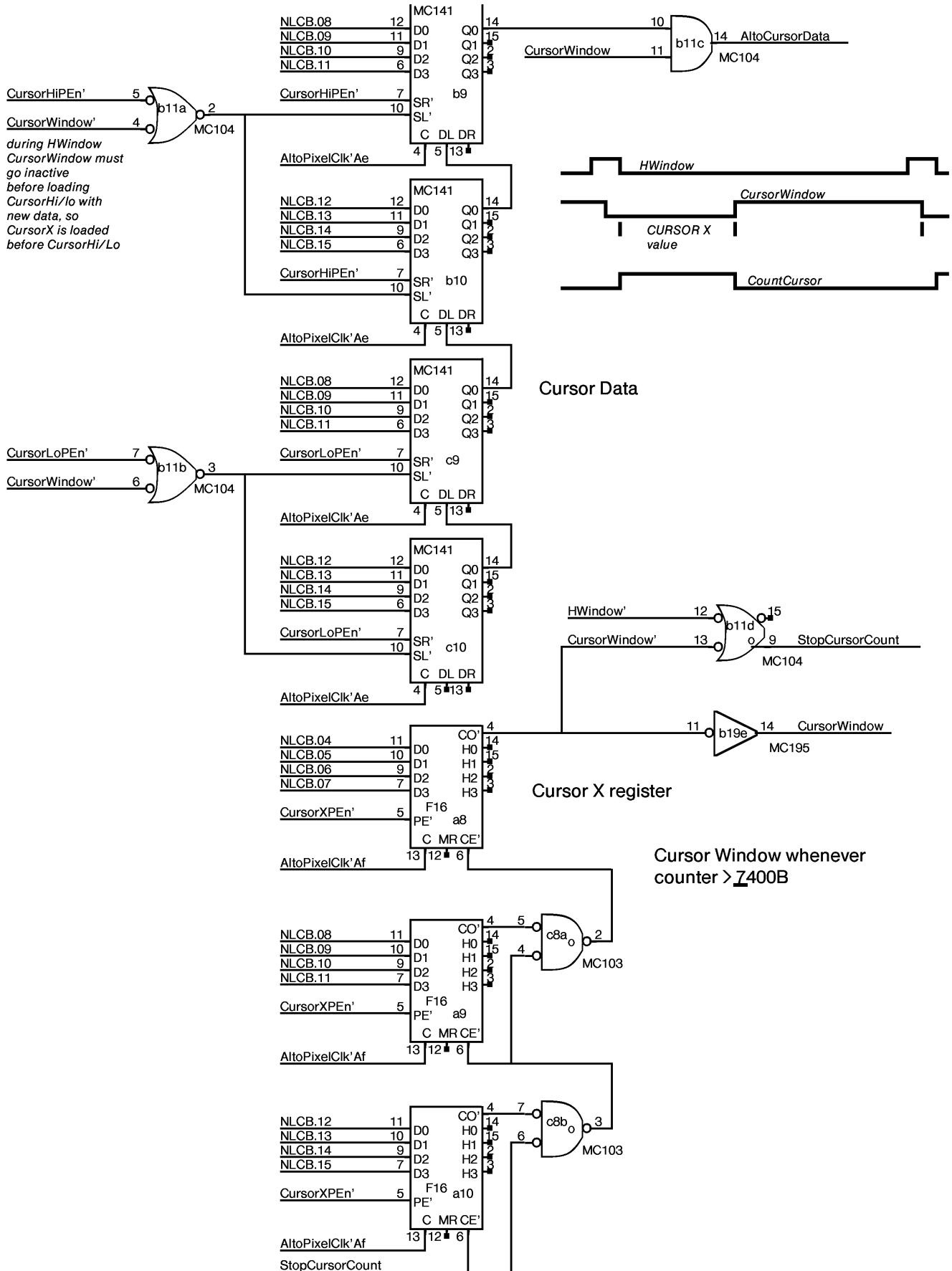
WakeAWT: (CurrentWCBFlag) OR (NextWCBFlag AND NOT(CurrentWCBFlag))

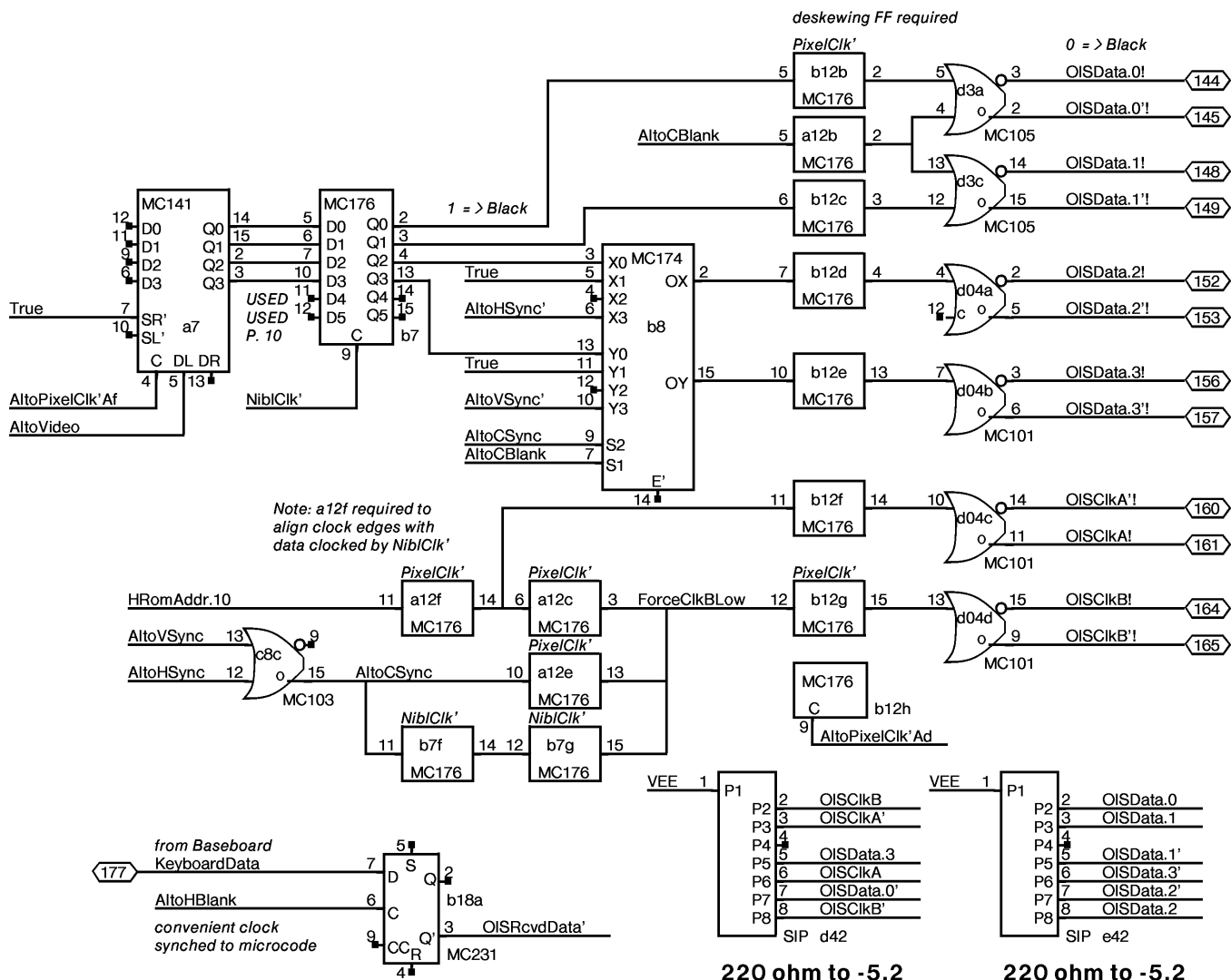
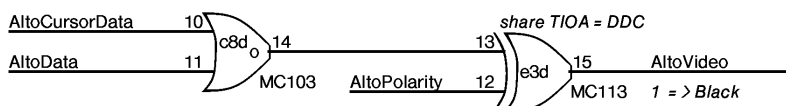
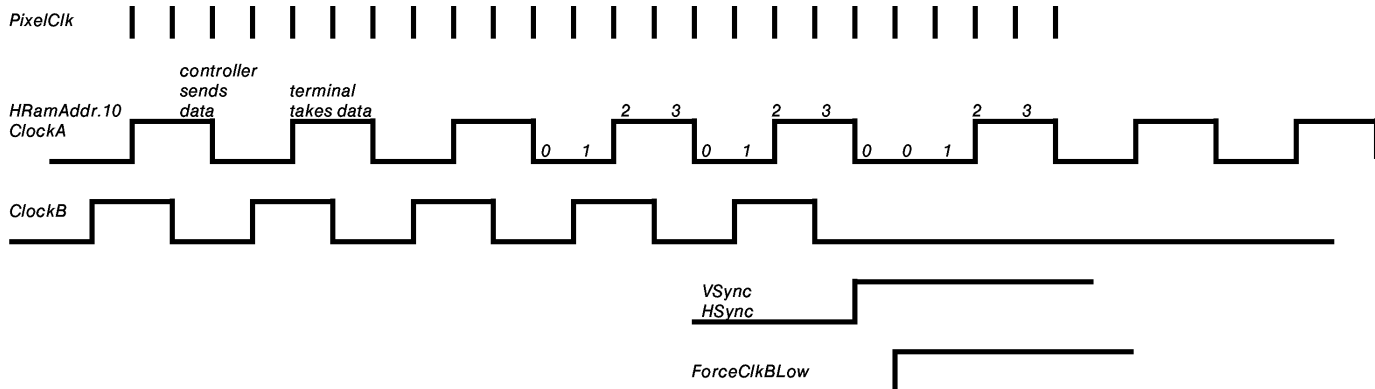
Flag management	SET	CLEARED
NextWCBFlag	by AHT when it has filled the NextWCB	by AWT when it has copied NextWCB into CurrentWCB
CurrentWCBFlag	by AWT when it has copied NextWCB into CurrentWCB	by AWT when it has sent out all the data from the CurrentWCB





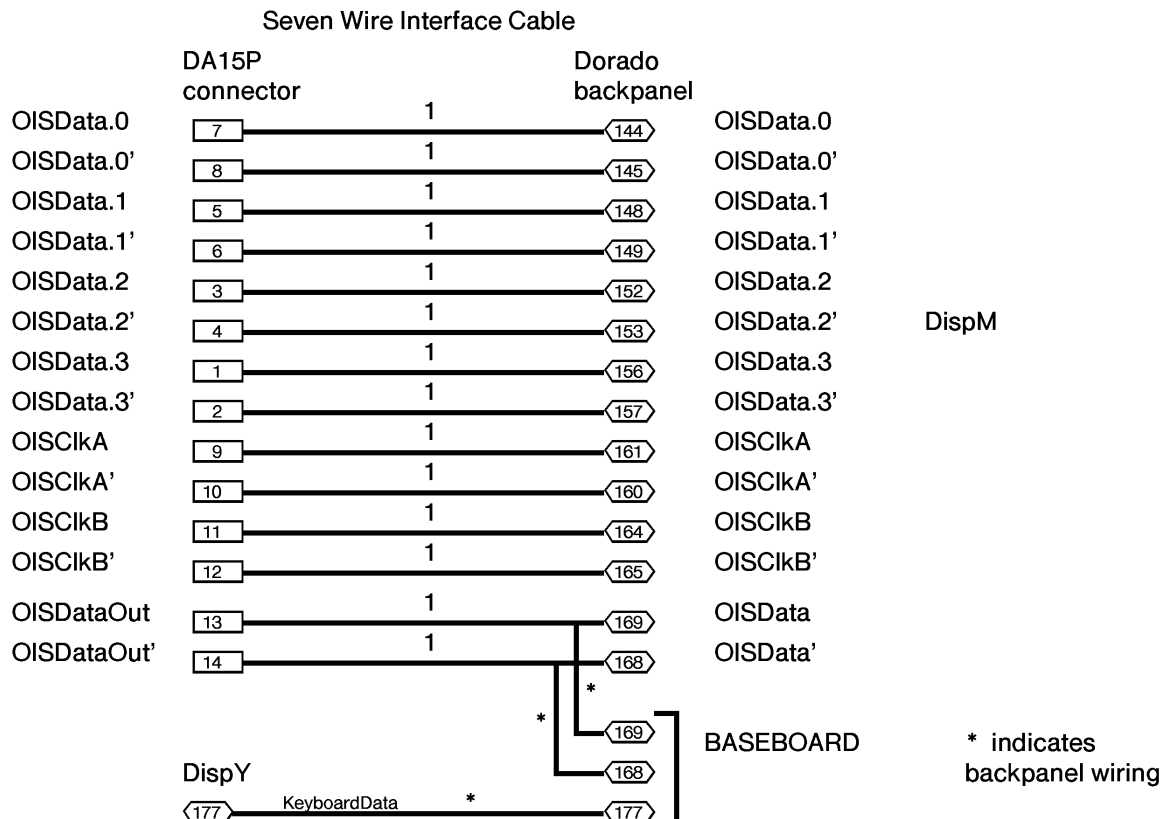
For AWT Task = 9D = 11B
cut legs 3 and 4

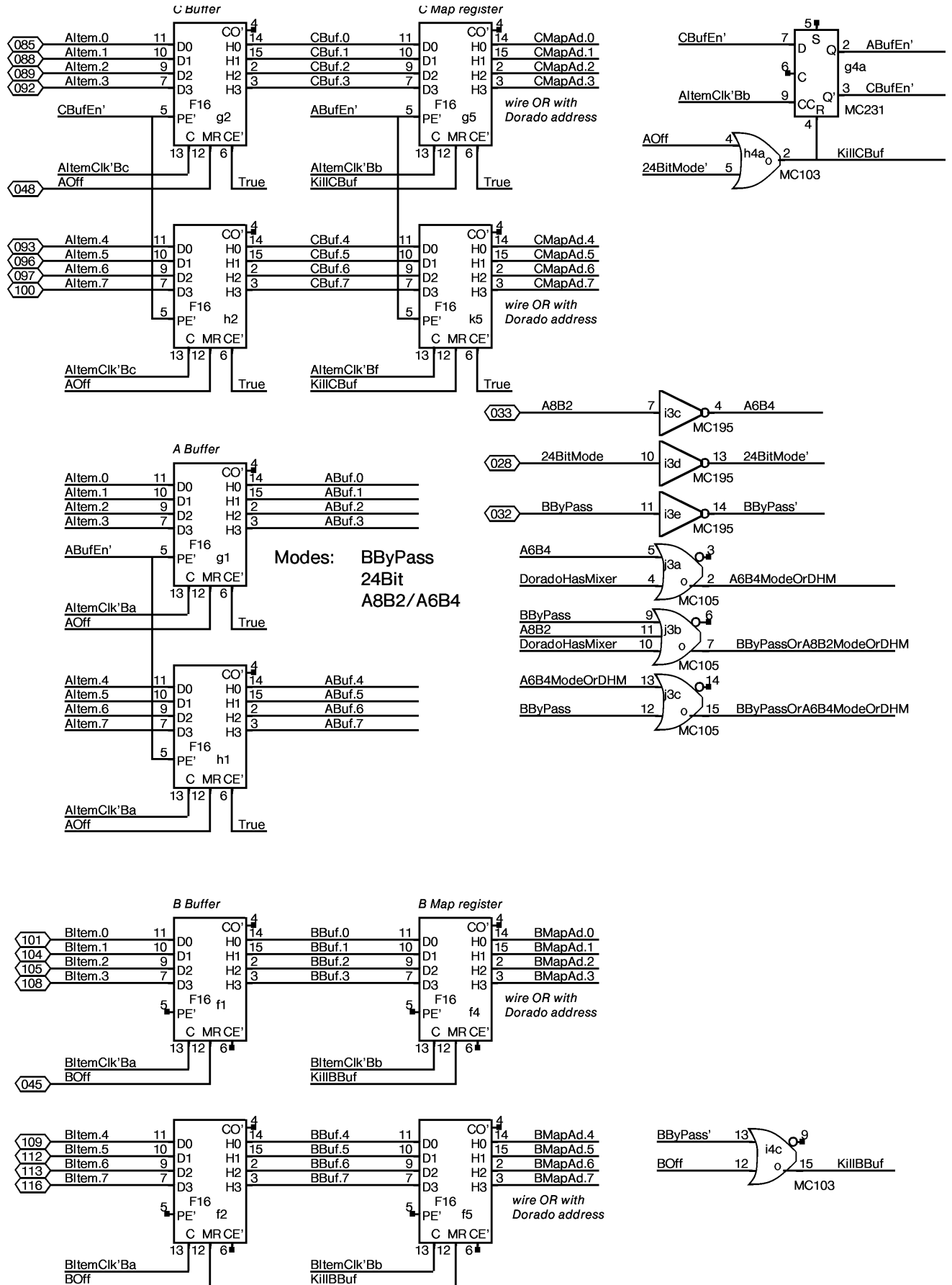




Alto Display Controller Next Line Control Block Format

Address	Name	Format
0	VCW vertical control word	0,..0, VBlank, VSync, OddField
1	Margin	LMarg[00..11]
2	Width	Width[00..11]
3	FifoAddr	FifoAddr[0..7] *must be even
4	Scan	AltoPolarity,0,0,0,0,0,0,
15	CursorX	CursorXCount[0..11]
16	CursorLo	CursorLoByte[4..11]
17	CursorHi	CursorHiByte[4..11]





BMapAd registers			
RIOB.08	11	F16 f6b	14 BMapAd.0
RIOB.09	10	F16 f6c	15 BMapAd.1
RIOB.10	9	F16 f6d	2 BMapAd.2
RIOB.11	7	F16 f6e	3 BMapAd.3
RIOB.12	11	F16 f7b	14 BMapAd.4
RIOB.13	10	F16 f7c	15 BMapAd.5
RIOB.14	9	F16 f7d	2 BMapAd.6
RIOB.15	7	F16 f7e	3 BMapAd.7

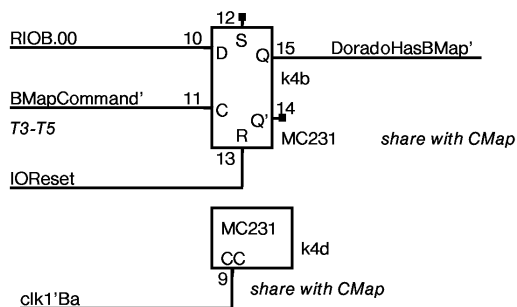
wire OR with
BMapAd registers

RIOB.08	19	MB071 i12b	23	ToBlue.0	wire OR with Mixer outputs
RIOB.09	18	MB071 i12c	22	ToBlue.1	
RIOB.10	17	MB071 i12d	21	ToBlue.2	
RIOB.11	16	MB071 i12e	20	ToBlue.3	
RIOB.12	19	MB071 j12b	23	ToBlue.4	
RIOB.13	18	MB071 j12c	22	ToBlue.5	
RIOB.14	17	MB071 j12d	21	ToBlue.6	
RIOB.15	16	MB071 j12e	20	ToBlue.7	
					wire OR with Mixer outputs

[illegible]

BMapAd.0	7	A0	MB071
BMapAd.1	8	A1	
BMapAd.2	9	A2	
BMapAd.3	10	A3	
BMapAd.4	11	A4	
BMapAd.5	12	A5	
BMapAd.6	13	A6	
BMapAd.7	14	A7	
BByPass'	6	BS0'	WE'
	5	BS1'	
	4	BS2'	
	3	BS3'	
WriteBMapClk1'a		2	

BMapAd.0	7	A0	MB071
BMapAd.1	8	A1	
BMapAd.2	9	A2	
BMapAd.3	10	A3	
BMapAd.4	11	A4	j12f
BMapAd.5	12	A5	
BMapAd.6	13	A6	
BMapAd.7	14	A7	
BByPass'	6	BS0'	WE'
	5	BS1'	
	4	BS2'	
	3	BS3'	
WriteBMapClk1'b		2	



NOTE: *in order to read/write BMap, BByPass mode must be ON and the B Channel must be OFF. This is inconvenient, but it does the right thing for real time mode switching.*

Dorado CMap address

wire OR with
CMapAd registers

RIOB.08	11	F16	14	CMapAd.0
		h5b		
RIOB.09	10	F16	15	CMapAd.1
		h5c		
RIOB.10	9	F16	2	CMapAd.2
		h5d		
RIOB.11	7	F16	3	CMapAd.3
		h5e		
RIOB.12	11	F16	14	CMapAd.4
		l5b		
RIOB.13	10	F16	15	CMapAd.5
		l5c		
RIOB.14	9	F16	2	CMapAd.6
		l5d		
RIOB.15	7	F16	3	CMapAd.7
		l5e		

wire OR with
CMapAd registers

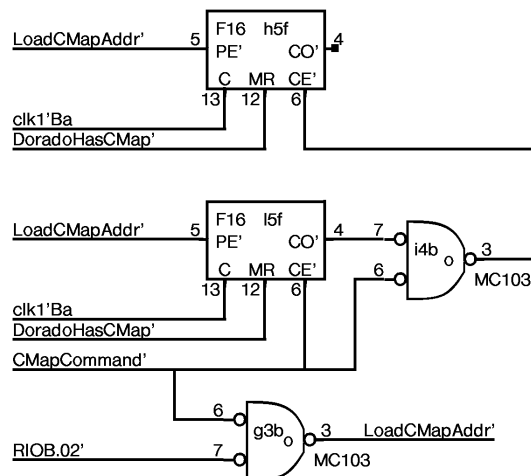
C Map

RIOB.08	19	MB071	23	ToGreen.0
		k14b		
RIOB.09	18	MB071	22	ToGreen.1
		k14c		
RIOB.10	17	MB071	21	ToGreen.2
		k14d		
RIOB.11	16	MB071	20	ToGreen.3
		k14e		
RIOB.12	19	MB071	23	ToGreen.4
		l14b		
RIOB.13	18	MB071	22	ToGreen.5
		l14c		
RIOB.14	17	MB071	21	ToGreen.6
		l14d		
RIOB.15	16	MB071	20	ToGreen.7
		l14e		

wire OR with
Mixer outputs

wire OR with
Mixer outputs

Dorado CMap address

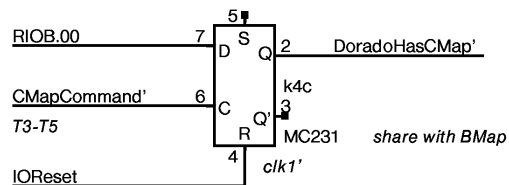


C Map

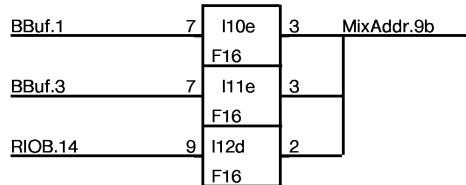
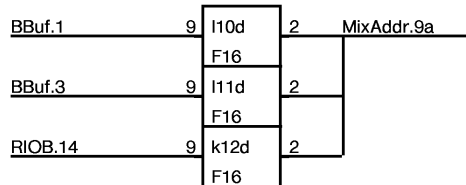
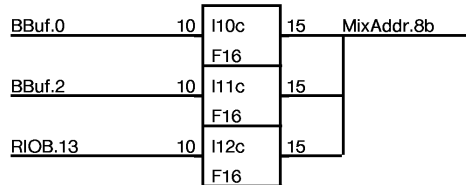
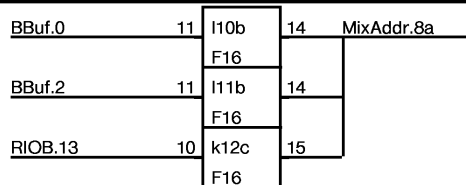
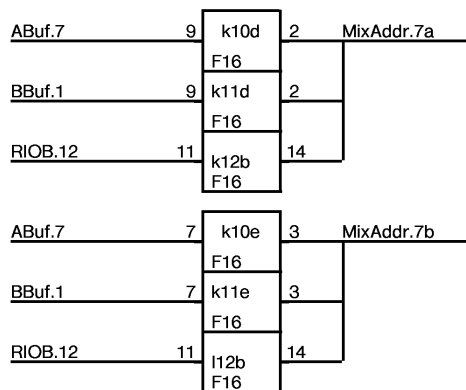
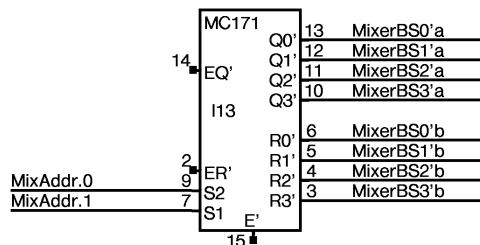
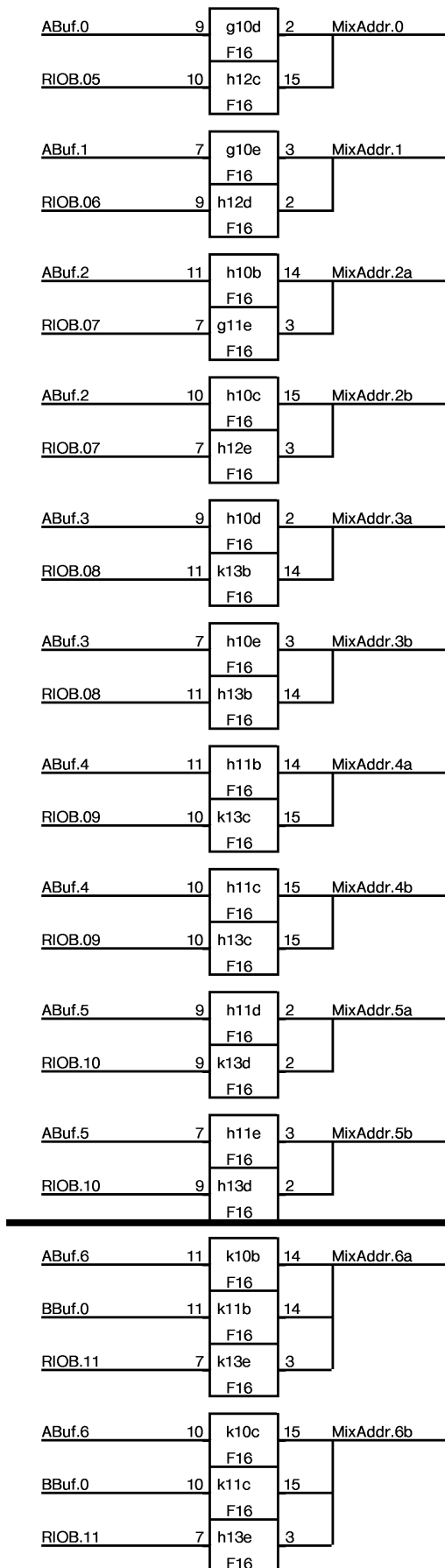
CMapAd.0	7	A0	MB071
CMapAd.1	8	A1	
CMapAd.2	9	A2	
CMapAd.3	10	A3	k14f
CMapAd.4	11	A4	
CMapAd.5	12	A5	
CMapAd.6	13	A6	
CMapAd.7	14	A7	
24BitMode'	6	BS0'	
	5	BS1'	
	4	BS2'	
	3	BS3'	
WriteCMapClk1'a	2	WE'	

C Map

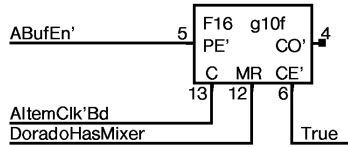
CMapAd.0	7	A0	MB071
CMapAd.1	8	A1	
CMapAd.2	9	A2	
CMapAd.3	10	A3	l14f
CMapAd.4	11	A4	
CMapAd.5	12	A5	
CMapAd.6	13	A6	
CMapAd.7	14	A7	
24BitMode'	6	BS0'	
	5	BS1'	
	4	BS2'	
	3	BS3'	
WriteCMapClk1'b	2	WE'	



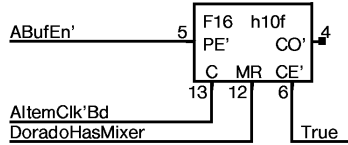
NOTE: in order to read/write CMap,
24BitMode must be ON and
The A channel must be OFF.
This is inconvenient, but it
does the right thing for
real time mode switching.



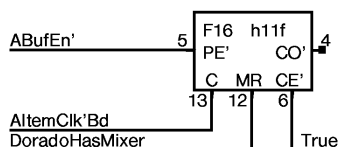
A Spare, Spare, 0, 1



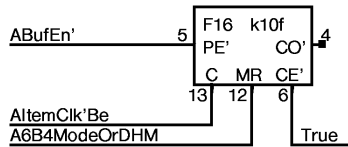
B 2a, 2b, 3a, 3b



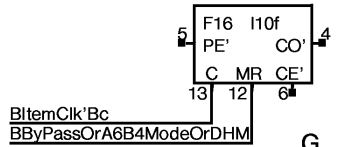
C 4a, 4b, 5a, 5b



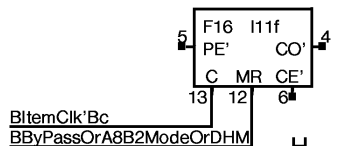
D 6a, 6b, 7a, 7b: Chan A



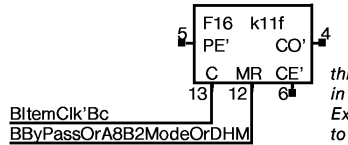
F 8a, 8b, 9a, 9b: Chan B



G 8a, 8b, 9a, 9b: Chan B

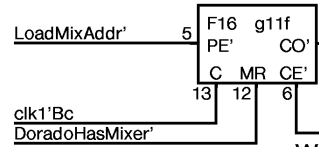


H 6a, 6b, 7a, 7b: Chan B

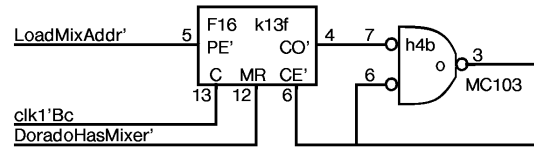


this must be OFF in 24Bit mode. Expect BByPass and A8B2 to be set whenever 24Bit mode is set.

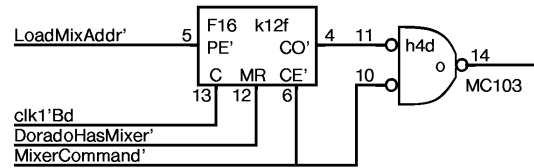
U , , , 2a



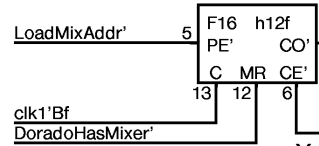
W 3a, 4a, 5a, 6a



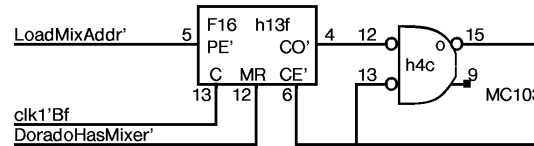
Y 7a, 8a, 9a, MSa



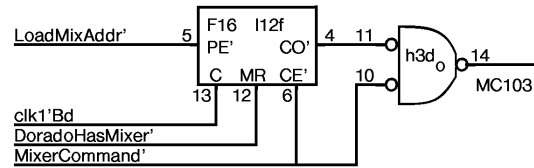
V Spare, 0, 1, 2b



X 3b, 4b, 5b, 6b

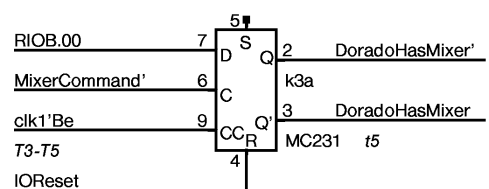
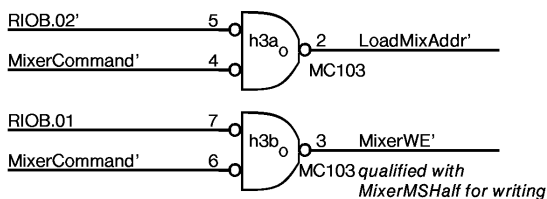


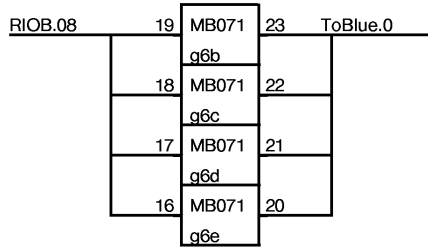
Z 7b, 8b, 9b, MSb



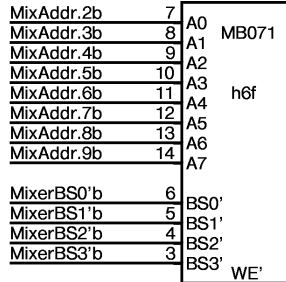
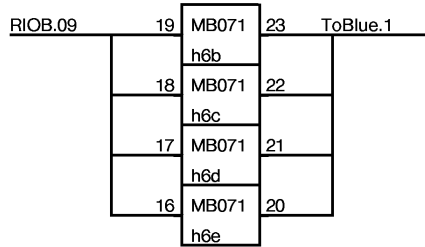
NOTE: B Channel must be started by microcode one pixel clk tick earlier than A channel for 24BitMode to align properly.

NOTE: Pixel clock must run at 2X rate for 24BitMode to work across entire screen !!

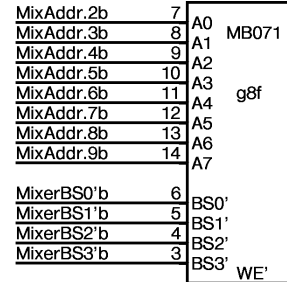
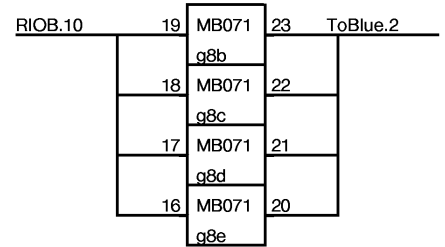




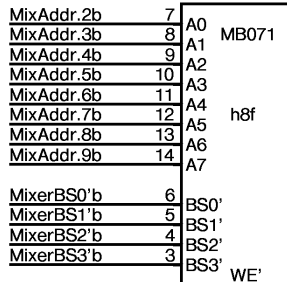
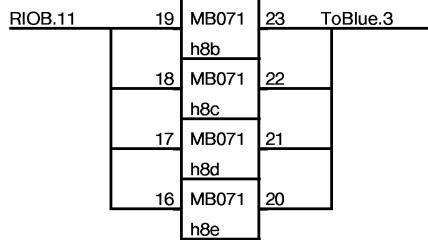
WriteMixerLo'Bb 2



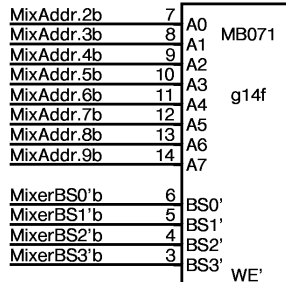
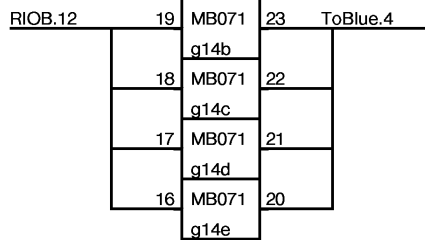
WriteMixerLo'Bb 2



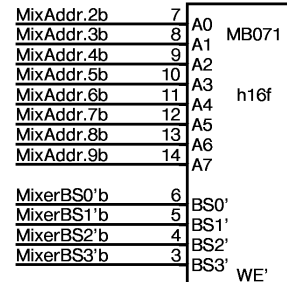
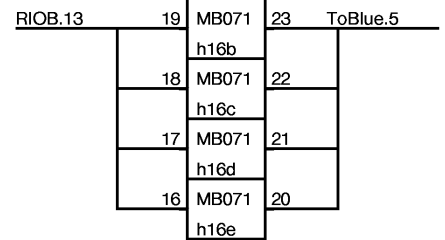
WriteMixerLo'Bc 2



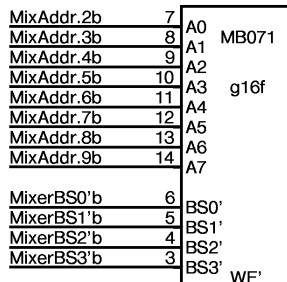
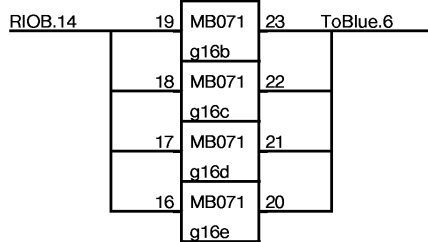
WriteMixerLo'Bc 2



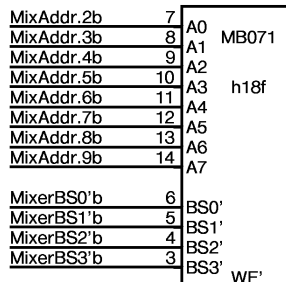
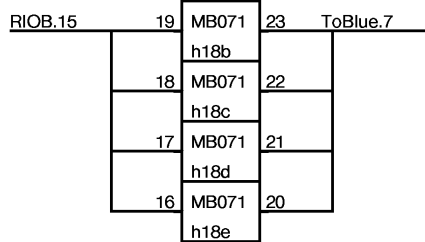
WriteMixerLo'Dc 2



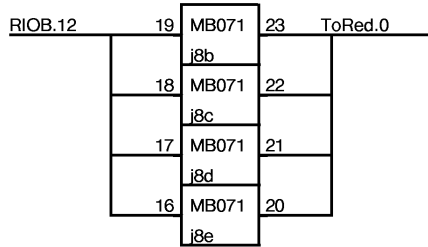
WriteMixerLo'Db 2



WriteMixerLo'Dc 2

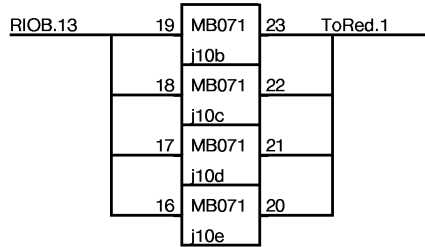


WriteMixerLo'Db 2



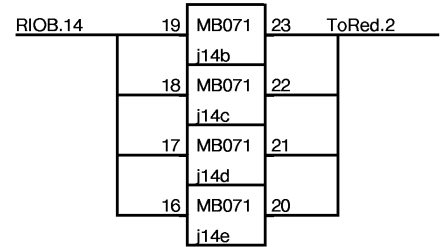
MixAddr.2a	7	A0
MixAddr.3a	8	A1 MB071
MixAddr.4a	9	A2
MixAddr.5a	10	A3
MixAddr.6a	11	A4
MixAddr.7a	12	A5 j8f
MixAddr.8a	13	A6
MixAddr.9a	14	A7
MixerBS0'a	6	BS0'
MixerBS1'a	5	BS1'
MixerBS2'a	4	BS2'
MixerBS3'a	3	BS3'

WriteMixerHi'Bc 2



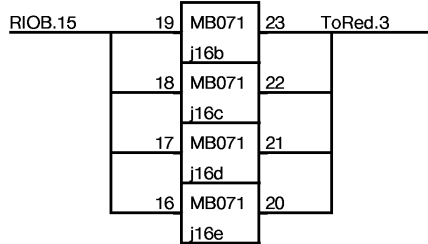
MixAddr.2a	7	A0
MixAddr.3a	8	A1 MB071
MixAddr.4a	9	A2
MixAddr.5a	10	A3
MixAddr.6a	11	A4
MixAddr.7a	12	A5 j10f
MixAddr.8a	13	A6
MixAddr.9a	14	A7
MixerBS0'a	6	BS0'
MixerBS1'a	5	BS1'
MixerBS2'a	4	BS2'
MixerBS3'a	3	BS3'

WriteMixerHi'Bc 2



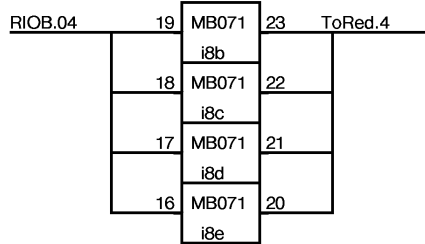
MixAddr.2a	7	A0
MixAddr.3a	8	A1 MB071
MixAddr.4a	9	A2
MixAddr.5a	10	A3
MixAddr.6a	11	A4
MixAddr.7a	12	A5 j14f
MixAddr.8a	13	A6
MixAddr.9a	14	A7
MixerBS0'a	6	BS0'
MixerBS1'a	5	BS1'
MixerBS2'a	4	BS2'
MixerBS3'a	3	BS3'

WriteMixerHi'Dc 2



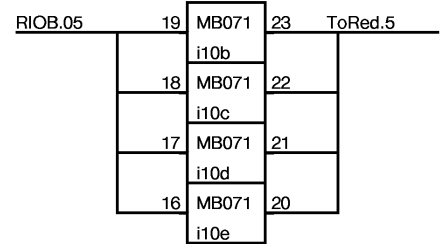
MixAddr.2a	7	A0
MixAddr.3a	8	A1 MB071
MixAddr.4a	9	A2
MixAddr.5a	10	A3
MixAddr.6a	11	A4 j16f
MixAddr.7a	12	A5
MixAddr.8a	13	A6
MixAddr.9a	14	A7
MixerBS0'a	6	BS0'
MixerBS1'a	5	BS1'
MixerBS2'a	4	BS2'
MixerBS3'a	3	BS3'

WriteMixerHi'Dc 2



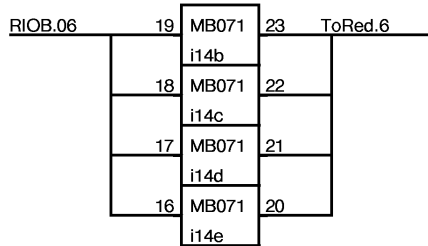
MixAddr.2b	7	A0
MixAddr.3b	8	A1 MB071
MixAddr.4b	9	A2
MixAddr.5b	10	A3 i8f
MixAddr.6b	11	A4
MixAddr.7b	12	A5
MixAddr.8b	13	A6
MixAddr.9b	14	A7
MixerBS0'b	6	BS0'
MixerBS1'b	5	BS1'
MixerBS2'b	4	BS2'
MixerBS3'b	3	BS3'

WriteMixerLo'Ba 2



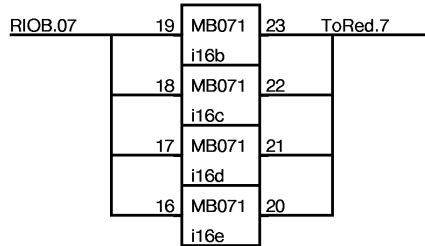
MixAddr.2b	7	A0
MixAddr.3b	8	A1 MB071
MixAddr.4b	9	A2
MixAddr.5b	10	A3 i10f
MixAddr.6b	11	A4
MixAddr.7b	12	A5
MixAddr.8b	13	A6
MixAddr.9b	14	A7
MixerBS0'b	6	BS0'
MixerBS1'b	5	BS1'
MixerBS2'b	4	BS2'
MixerBS3'b	3	BS3'

WriteMixerLo'Ba 2



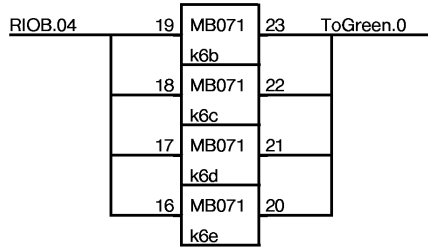
MixAddr.2b	7	A0
MixAddr.3b	8	A1 MB071
MixAddr.4b	9	A2
MixAddr.5b	10	A3 i14f
MixAddr.6b	11	A4
MixAddr.7b	12	A5
MixAddr.8b	13	A6
MixAddr.9b	14	A7
MixerBS0'b	6	BS0'
MixerBS1'b	5	BS1'
MixerBS2'b	4	BS2'
MixerBS3'b	3	BS3'

WriteMixerLo'Da 2



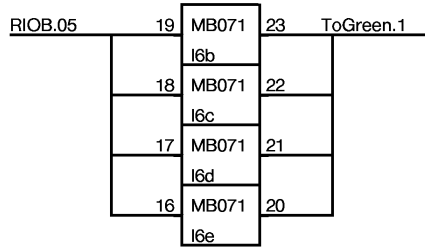
MixAddr.2b	7	A0
MixAddr.3b	8	A1 MB071
MixAddr.4b	9	A2
MixAddr.5b	10	A3 i16f
MixAddr.6b	11	A4
MixAddr.7b	12	A5
MixAddr.8b	13	A6
MixAddr.9b	14	A7
MixerBS0'b	6	BS0'
MixerBS1'b	5	BS1'
MixerBS2'b	4	BS2'
MixerBS3'b	3	BS3'

WriteMixerLo'Da 2



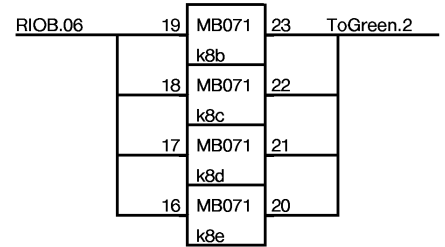
MixAddr.2a	7	A0	MB071
MixAddr.3a	8	A1	
MixAddr.4a	9	A2	
MixAddr.5a	10	A3	
MixAddr.6a	11	A4	k6f
MixAddr.7a	12	A5	
MixAddr.8a	13	A6	
MixAddr.9a	14	A7	
MixerBS0'a	6	BS0'	
MixerBS1'a	5	BS1'	
MixerBS2'a	4	BS2'	
MixerBS3'a	3	BS3'	
		WE'	

WriteMixerHi' Ba 2



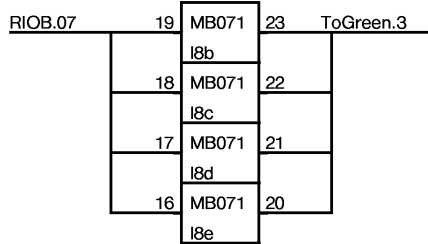
MixAddr.2a	7	A0	MB071
MixAddr.3a	8	A1	
MixAddr.4a	9	A2	
MixAddr.5a	10	A3	
MixAddr.6a	11	A4	
MixAddr.7a	12	A5	
MixAddr.8a	13	A6	
MixAddr.9a	14	A7	
MixerBS0'a	6	BS0'	l6f
MixerBS1'a	5	BS1'	
MixerBS2'a	4	BS2'	
MixerBS3'a	3	BS3'	
			WE'

WriteMixerHi' Bb 2



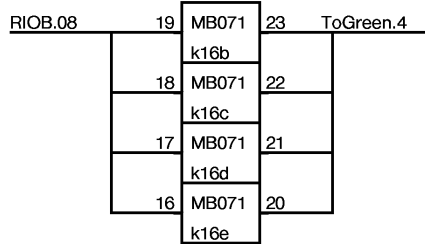
MixAddr.2a	7	A0	MB071
MixAddr.3a	8	A1	
MixAddr.4a	9	A2	
MixAddr.5a	10	A3	
MixAddr.6a	11	A4	
MixAddr.7a	12	A5	
MixAddr.8a	13	A6	
MixAddr.9a	14	A7	
MixerBS0'a	6	BS0'	k8f
MixerBS1'a	5	BS1'	
MixerBS2'a	4	BS2'	
MixerBS3'a	3	BS3'	
			WE'

WriteMixerHi' Ba 2



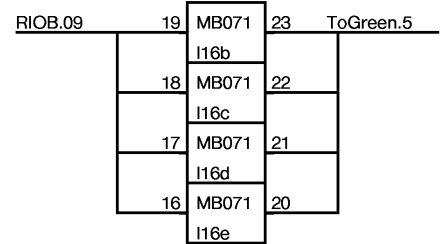
MixAddr.2a	7	A0	MB071
MixAddr.3a	8	A1	
MixAddr.4a	9	A2	
MixAddr.5a	10	A3	
MixAddr.6a	11	A4	
MixAddr.7a	12	A5	
MixAddr.8a	13	A6	
MixAddr.9a	14	A7	
MixerBS0'a	6	BS0'	l8f
MixerBS1'a	5	BS1'	
MixerBS2'a	4	BS2'	
MixerBS3'a	3	BS3'	
			WF'

WriteMixerHi' Bb 2



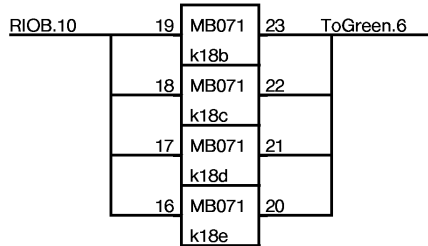
MixAddr.2a	7	A0	MB071
MixAddr.3a	8	A1	
MixAddr.4a	9	A2	
MixAddr.5a	10	A3	
MixAddr.6a	11	A4	
MixAddr.7a	12	A5	
MixAddr.8a	13	A6	
MixAddr.9a	14	A7	
MixerBS0'a	6	BS0'	k16f
MixerBS1'a	5	BS1'	
MixerBS2'a	4	BS2'	
MixerBS3'a	3	BS3'	
			WF'

WriteMixerHi' Da 2



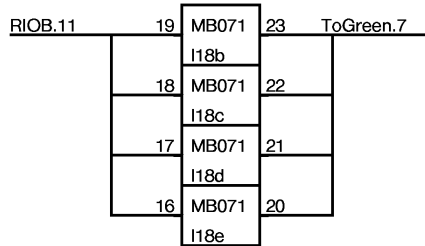
MixAddr.2a	7	A0	MB071
MixAddr.3a	8	A1	
MixAddr.4a	9	A2	
MixAddr.5a	10	A3	
MixAddr.6a	11	A4	
MixAddr.7a	12	A5	
MixAddr.8a	13	A6	
MixAddr.9a	14	A7	
MixerBS0'a	6	BS0'	l16f
MixerBS1'a	5	BS1'	
MixerBS2'a	4	BS2'	
MixerBS3'a	3	BS3'	
			WF'

WriteMixerHi' Da 2



MixAddr.2a	7	A0	MB071 k18f
MixAddr.3a	8	A1	
MixAddr.4a	9	A2	
MixAddr.5a	10	A3	
MixAddr.6a	11	A4	
MixAddr.7a	12	A5	
MixAddr.8a	13	A6	
MixAddr.9a	14	A7	
MixerBS0'a	6	BS0'	WF'
MixerBS1'a	5	BS1'	
MixerBS2'a	4	BS2'	
MixerBS3'a	3	BS3'	

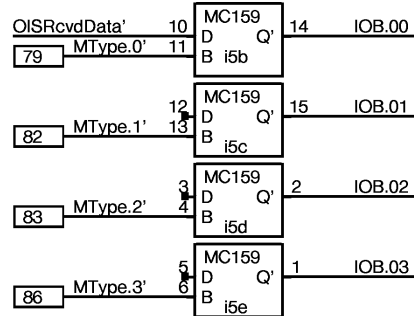
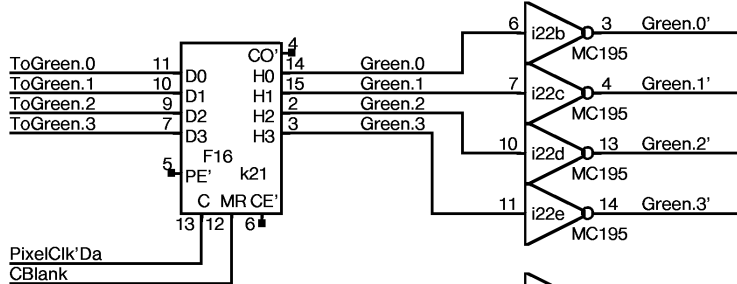
WriteMixerHi' Db 2



MixAddr.2a	7	A0	MB071
MixAddr.3a	8	A1	
MixAddr.4a	9	A2	
MixAddr.5a	10	A3	
MixAddr.6a	11	A4	
MixAddr.7a	12	A5	
MixAddr.8a	13	A6	
MixAddr.9a	14	A7	
MixerBS0'a	6	BS0'	l18f
MixerBS1'a	5	BS1'	
MixerBS2'a	4	BS2'	
MixerBS3'a	3	BS3'	
			WE'

WriteMixerHi' Db 2

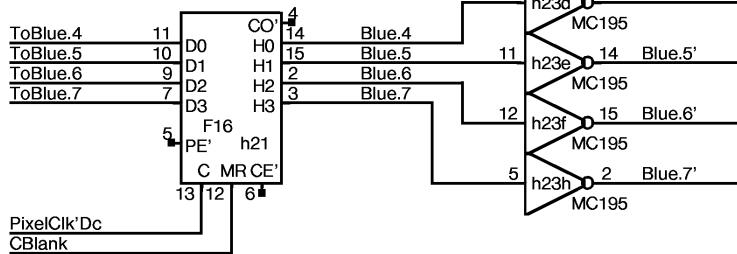
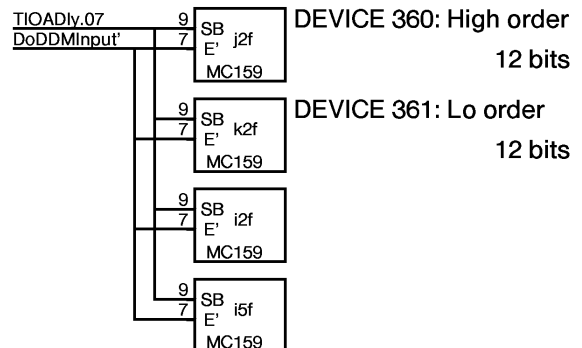
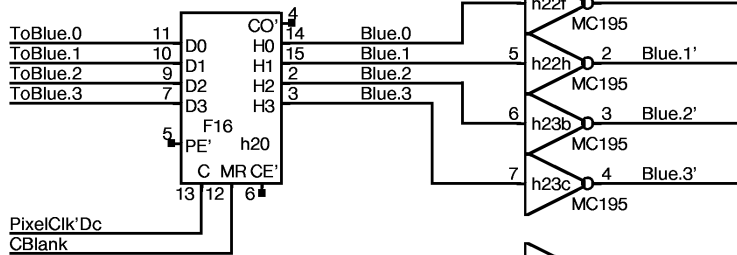
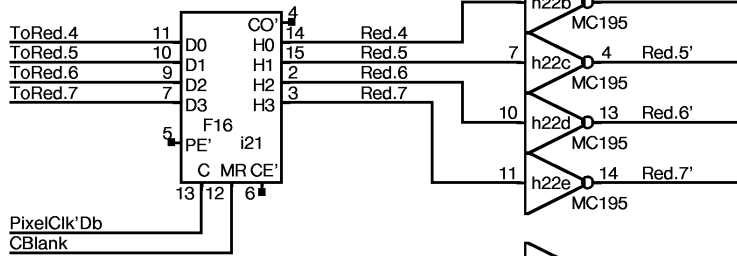
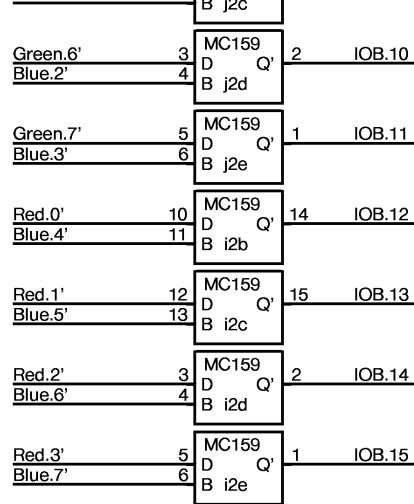
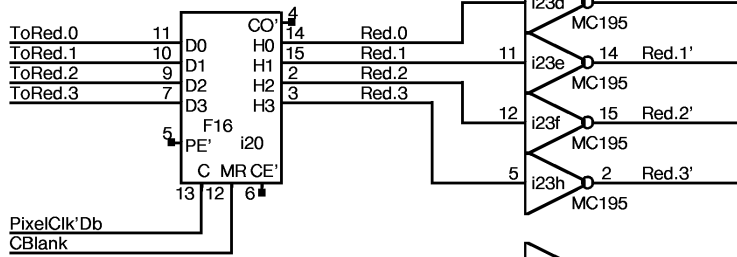
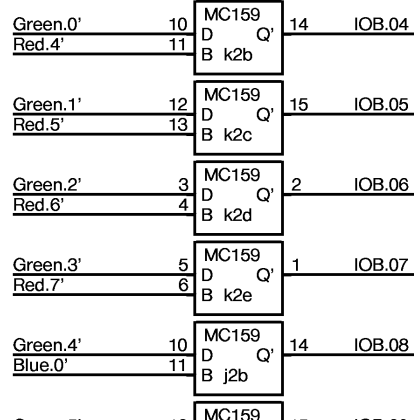
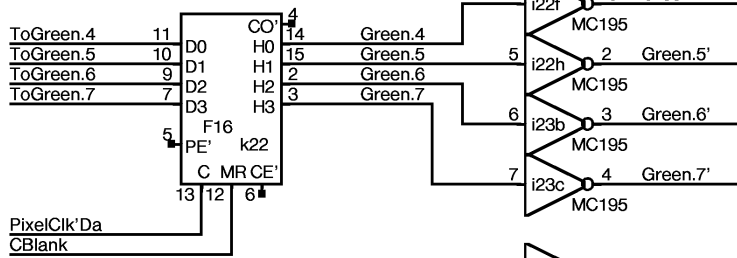
NO Parity supplied
Use InputNoPE

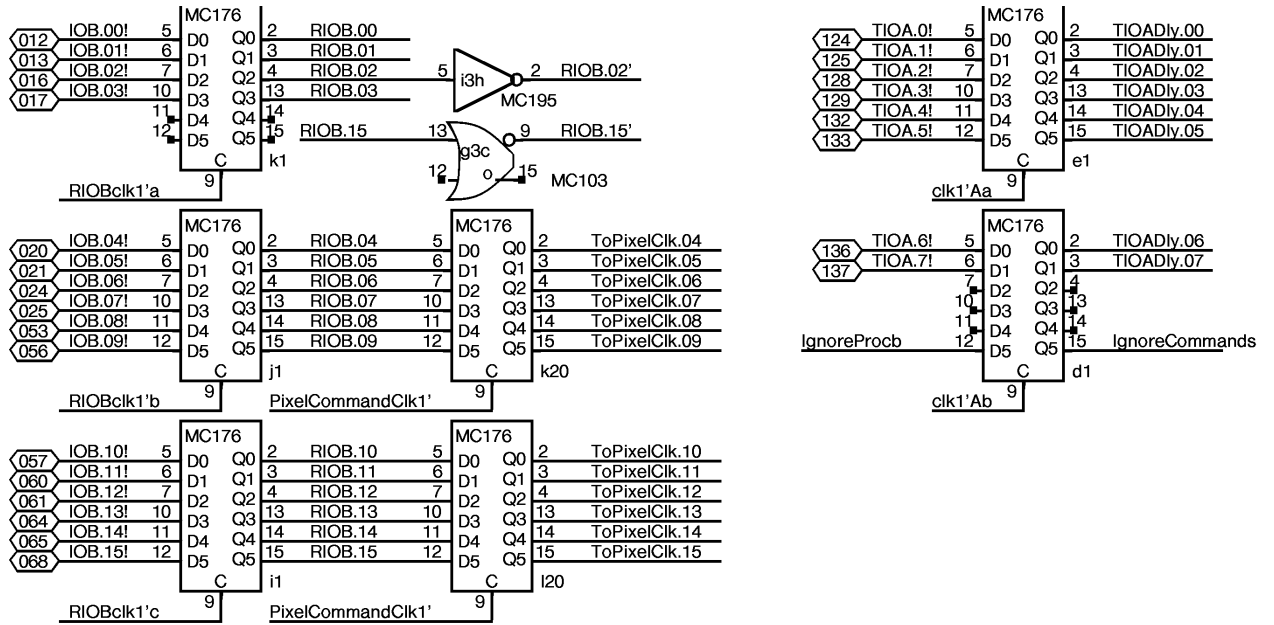


Device 360:
keyboard data

Device 360:
Here I Am!!

Device 361:
Monitor type code
Jumped on
backplane

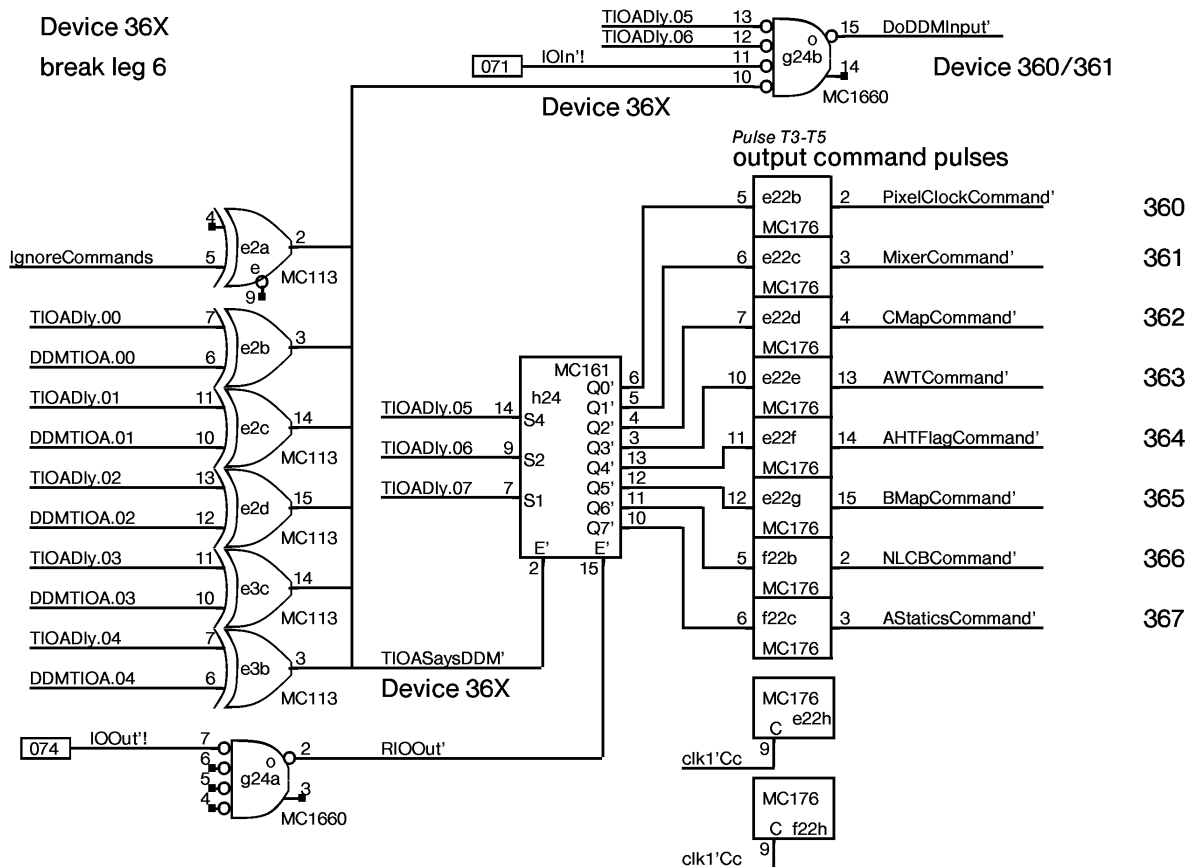




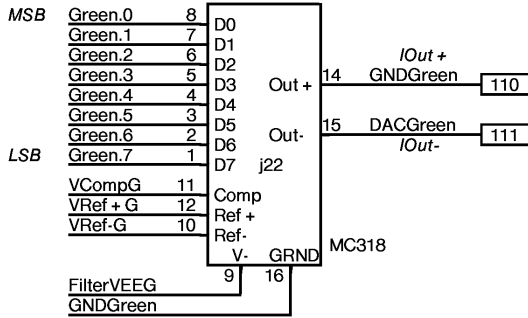
True 1 SIP b g41b
DDMTIOA.00 1 2 SIP c g41c
DDMTIOA.01 1 3 SIP d g41d
DDMTIOA.02 1 4 SIP e g41e
DDMTIOA.03 1 5 SIP f g41f
DDMTIOA.04 0 6 SIP g g41g

Device 36X

break leg 6

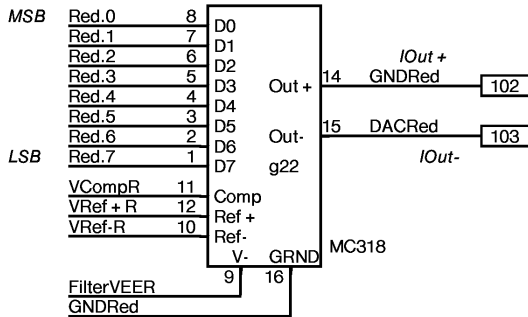


NOTES:
GNDGreen is
used as single point GND
for DAC system



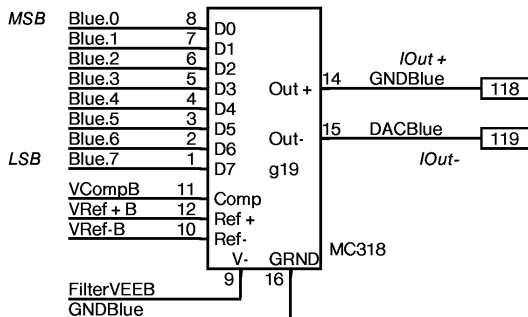
Digital/Analog converter

NOTES:
GND RED is
used as single point GND
for DAC system

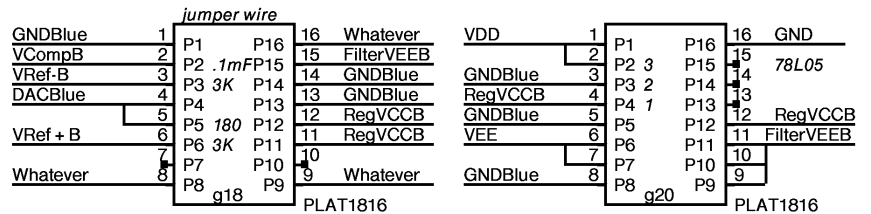
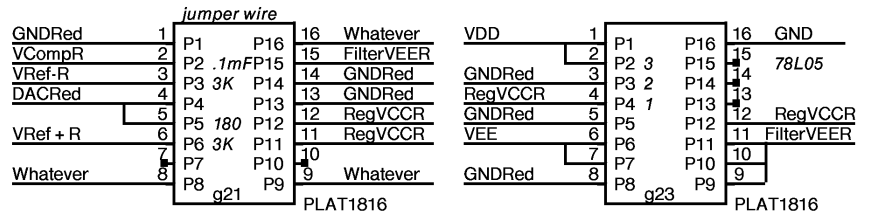
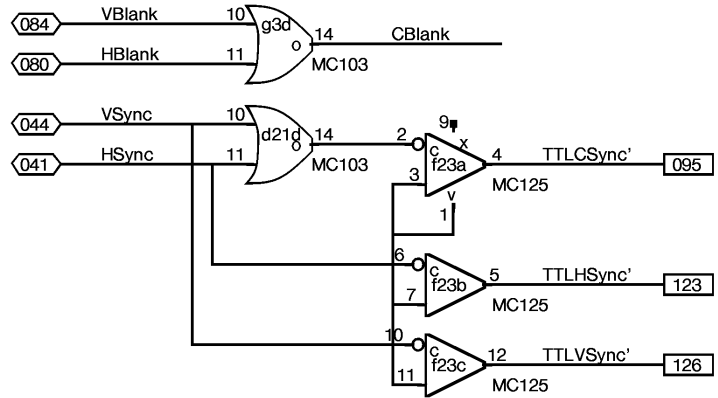
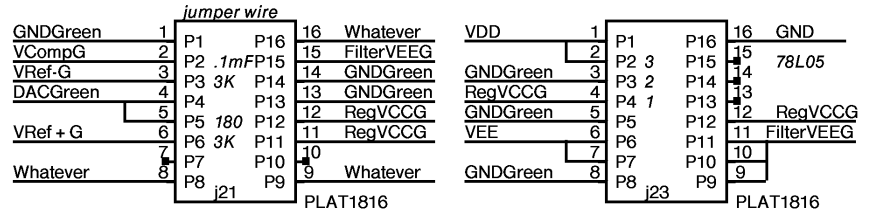


Digital/Analog converter

NOTES:
GNDBLue is
used as single point GND
for DAC system



Digital/Analog converter

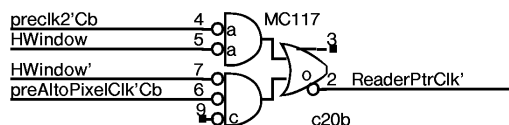
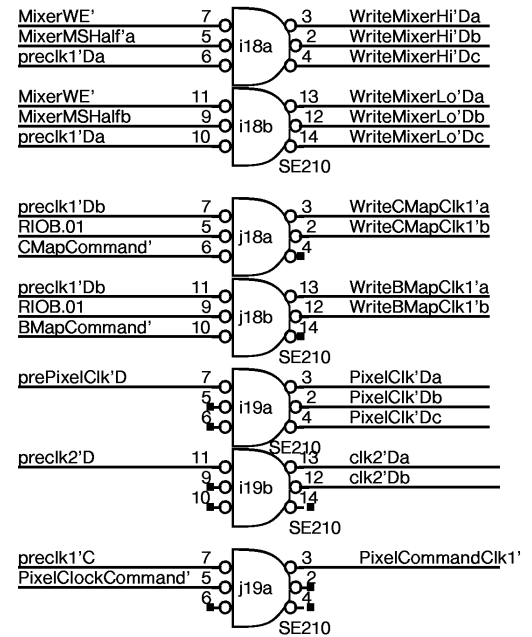
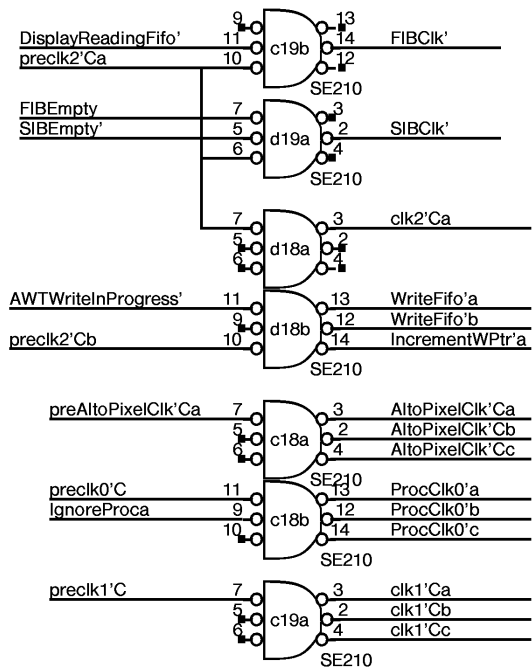
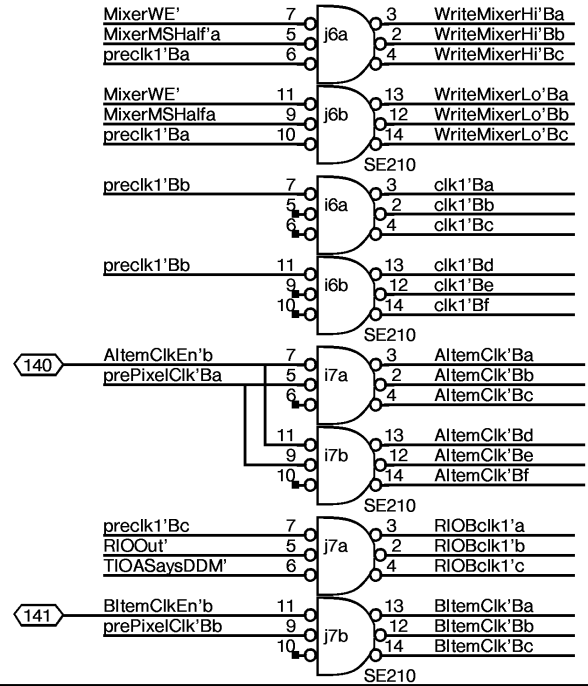
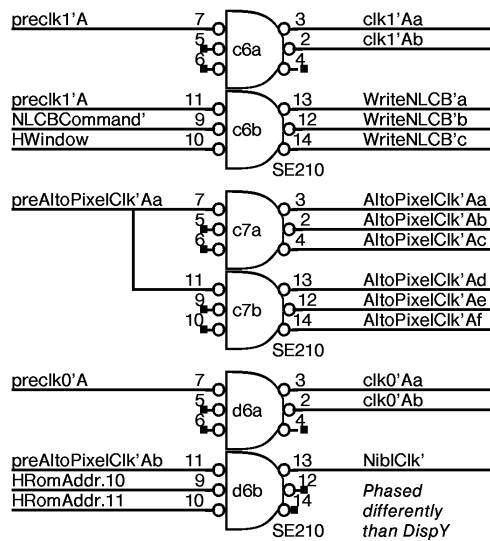


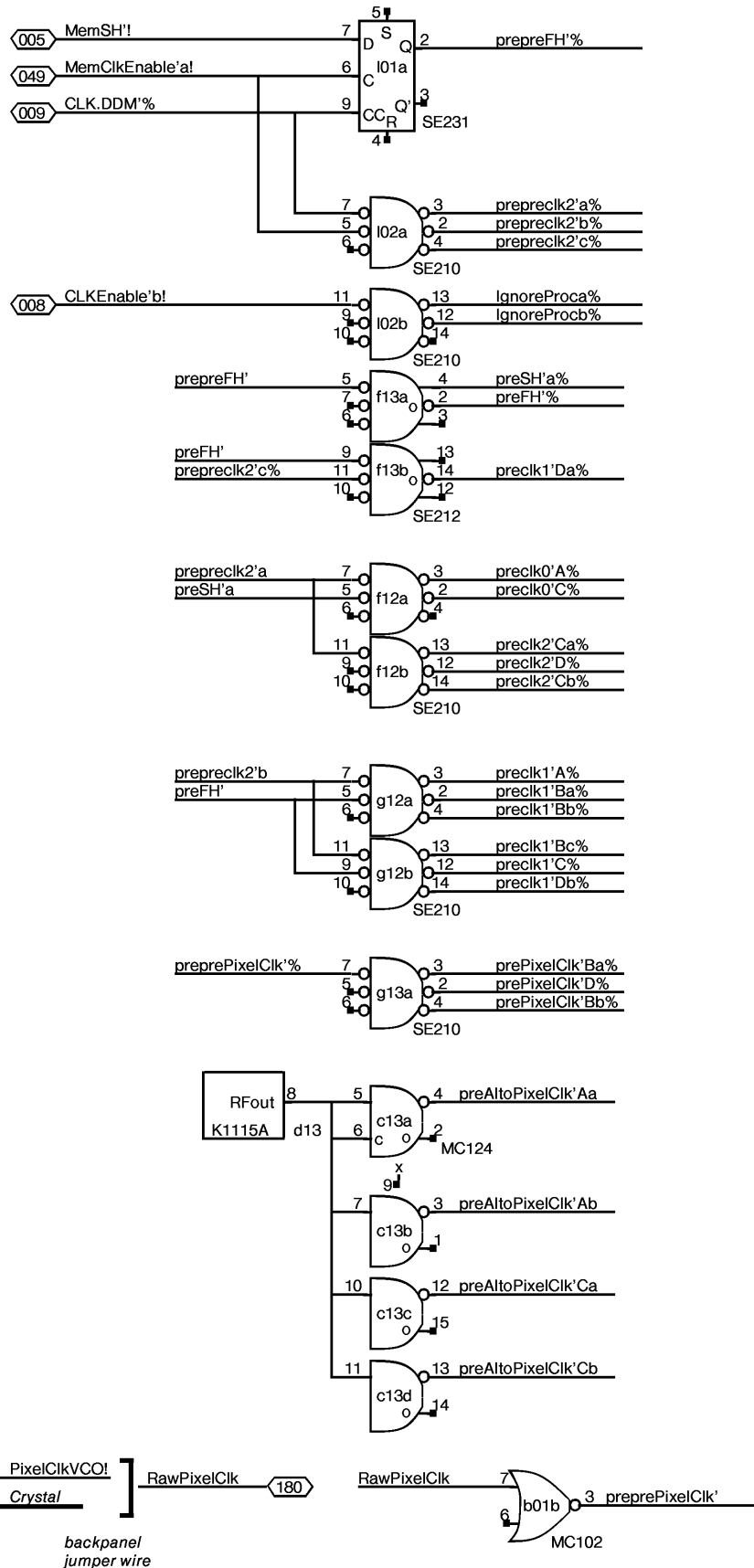
1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4		13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

1	0.1mFarad	16	
2	in ³	78L05	15
3	gnd ²		14
4	out ¹		13
5	0.1mFarad		12
6	12 mHnry		11
7	2 ohm		10
8	+ 22 mF		09

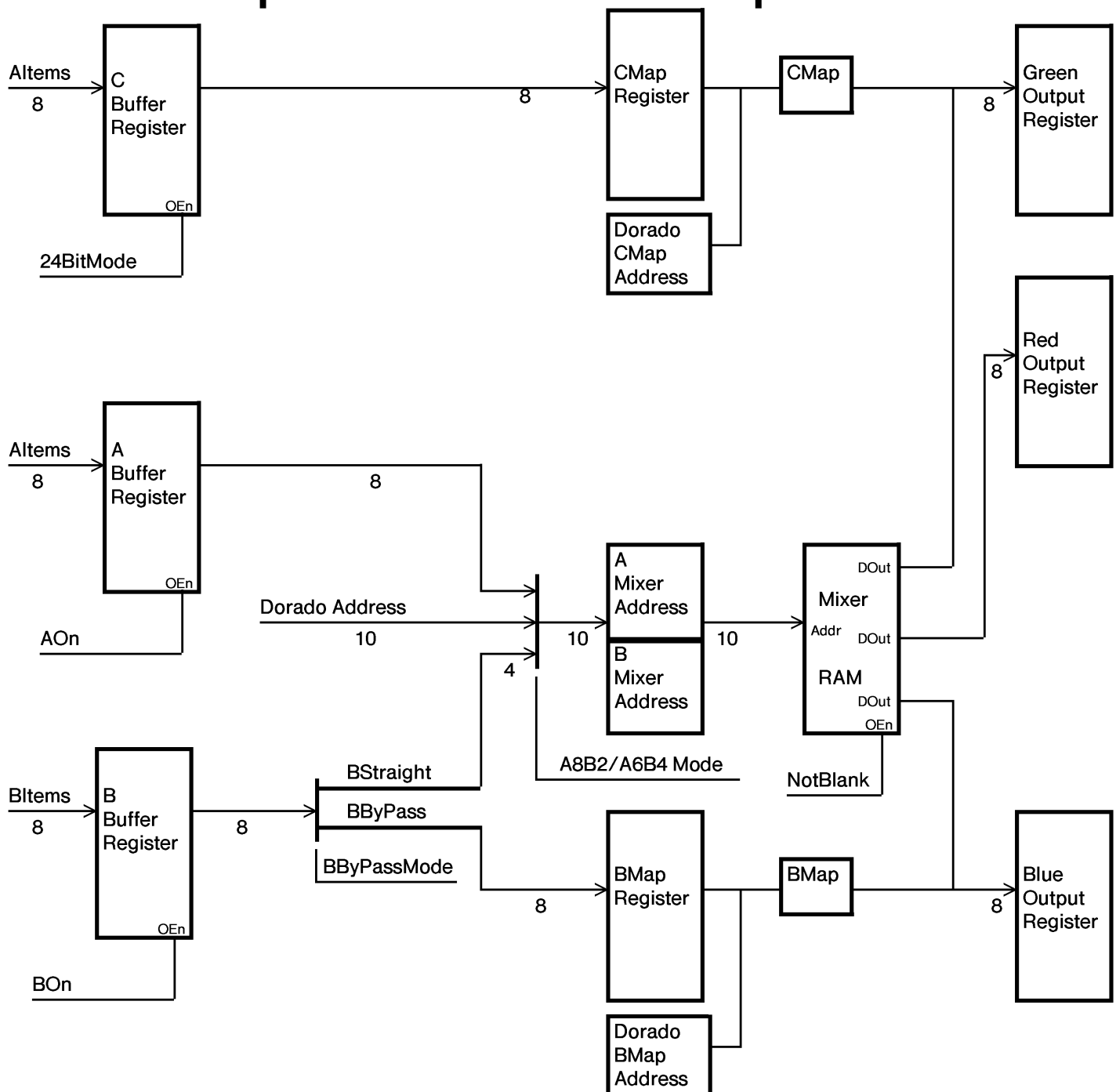
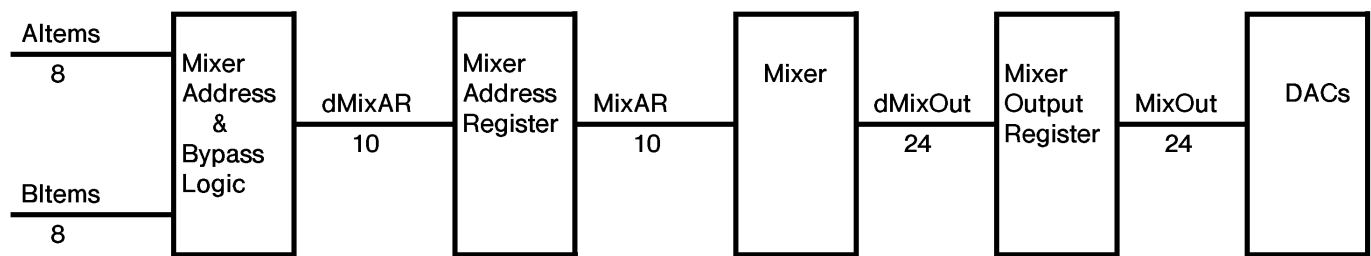
locations g18,g21,j21

locations j23,g23,g20

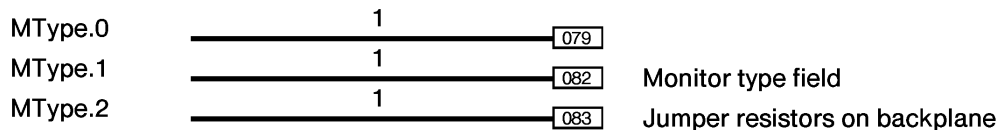
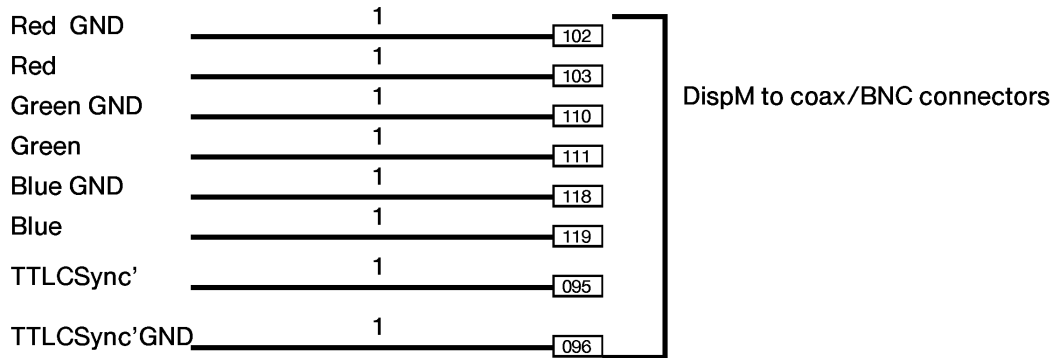




		CONNECTOR				CONNECTOR				IOBLO		CONNECTOR		IOBHI		CLK		
A	B>C	a 181	b 168	c 153	d 137	e 124	f 109	93 g	80 h	64 i	48 j	33 k	20 l					B
1 41	DISC VCO	PCLK 8,25,23,23 102			TIOA 176 21	TIOA 176 21	BBuf F16 12	ABuf F16 12	ABuf F16 12	RIOB 176 21	RIOB 176 21	RIOB 176 21	25 SE231					1
2	DISC VCO	12040 VCO			WakeAHT 135	TIOA = D 113	BBuf F16 12	CBuf F16 12	CBuf F16 12	DIOB 159 20	DIOB 159 20	DIOB 159 20	25 SE210					2
3 42	3140 VCO	136 VCO	DISC		OISDrvr 10,4,10 105	TIOA = D A,21,21,10 113	WakeAWT AVSync 231	MapLd 13,14,21, 103 22	MixLd 16,16,20, 103 16	Modes B,12,12 195 12,15,2	DHM 105 12	DHMix 231 16,B					3	
4	DISC VCO	95016 VCO	VCO		OISDrvr 101 10		BMapAd F16 12	ABufEn 231 12,B	ORs 12,16,16, 103 16	13,14,12,D 103		DHMap 231 13,14					4	
5 43	1648 VCO	95016 VCO	K1115 VCO		LS109 VCO		BMapAd F16 12	CMapAd F16 12	DCMapAd F16 14	IOB 159 20		CMapAd F16 12	DCMapAd F16 14					5
6	DISC VCO	95016 VCO	Clock		Clock	ShutUp 231 4	DBMapAd F16 13	Mixer B0 17	Mixer B1 17	Clock 210 24	Clock 210 24	Mixer G0 19	Mixer G1 19					6
7 44	AltoStoP	Nibble	Clock			CLCBDCd 161 5	DBMapAd F16 13	--	--	Clock 210 24	Clock 210 24	--	--					7
8	CursorX F16 9	OIS 174 10	OIS 9,9,10,10 103		Width F16 4	LMarg F16 4	NLCB 145A 5	Mixer B2 17	Mixer B3 17	Mixer R4 18	Mixer R0 18	Mixer G2 19	Mixer G3 19					8
9 45	CursorX F16 9	CursorHi 141 9	CursorLo 141 9		Width F16 4	LMarg F16 4	NLCB 145A 5	--	--	--	--	--	--					9
10	CursorX F16 9	CursorHi 141 9	CursorLo 141 9		Width F16 4	LMarg F16 4	NLCB 145A 5	MixAddr F16 15	MixAddr F16 15	Mixer R5 18	Mixer R1 18	MixAddr F16 15	MixAddr F16 15					10
11 46	VCW F16 4	Cursor 104 9	Width 4,4,6,4 103		LMarg 103 4	NLCBAddr F16 5	NLCBAddr 197 5	MixAddr F16 15	MixAddr F16 15	--	--	MixAddr F16 15	MixAddr F16 15					11
12	PixelClk 10,10,6 176 10,F,6	OISSkew 176 10	3,4,4,2 104		HWindow 135 5	LoadASR 121 3	Clock 210 25	Clock 210 25	MixAddr F16 15	BMap 13	BMap 13	MixAddr F16 15	MixAddr F16 15					12
13 47	HRomAddr 6	HRomOut 176 6	TTLtoEcl 25		Xtal K1115 25	LoadASR F16 3	Clock 212 25	Clock 210 25	MixAddr F16 15	--	--	MixAddr F16 15	BSel 171 15					13
14	HRomAddr 6	ASR 141 3	ASR 141 3		ASR 141 3	ASR 141 3		Mixer B4 17		Mixer R6 18	Mixer R2 18	CMap 19	CMap 19					14
15 48	HRomAddr 6		FIB 176 3		FIB 176 3	FIB 176 3		--		--	--	--	--					15
16	HRom MC149 6	Fifo 3	Fifo 3		Fifo 3	Fifo 3	FifoAd 158 2	Mixer B6 17	Mixer B5 17	Mixer R7 18	Mixer R3 18	Mixer G4 19	Mixer G5 19					16
17 49		--	--		--	--	FifoAd 158 2	--	--	--	--	--	--					17
18	6,23,4,2 102	OISRcvd 231 10,B	Clock 210 24		Clock 210 24	RPtr 2	WPTr 2	BIDac Plat 22	Mixer B7 17	Clock 210 24	Clock 210 24	Mixer G6 19	Mixer G7 19					18
19 50	6,4,6,6 103	6,4,6 9,4,6 195	Clock 210 24		Clock 210 24,B	RPTr 2	WPTr 2	BIDac 10318	--	Clock 210 24	Clock 210 24	--	--					19
20		HRom MC149 6	RdrPtrClk CTisAWT 117 24,8		ASRSync 231 2		AOn ASRSync 135 2,4	BIDac Plat 22	Blue 20	Red 20		PXLCLK 176 21	PXLCLK 176 21					20
21 51			SCAN F16 4		7,2,2,22 103	Flags 231 7	Flags 118 7	RdDac Plat 22	Blue 20	Red 20	GrDac Plat 22	Green 20						21
22	SIB 176 3	SIB 176 3	SIB 176 3			CMND 176 21	CMND 176 21	RdDac 10318	Color' 195 20	Color' 195 20	GrDac 10318	Green 20						22
23 52	8,8,C 105	ProcClk0 176 8			blocked 135 8	ASRSync 135 2	TTLCSync 22 125	RdDac Plat 22	Color' 195 20	Color' 195 20	GrDac Plat 22		1,B,C,D 102					23
24	NextAWT 113 8				WakeCnt B,4*8,2 195	WakeCnt F16 8	Hold 231 8	IOInOut 1660 21	TIOA = D 161 21	Fout 176 1	Fout 176 1	Fout 176 1	FtTsk 113 1					24
C		a 11	b 26	c 39	d 55	e 70	Hold f 86	99 g	114 h	129 i	143 j	159 k	174 l	D				
E		Next	SubT	BLK	FIN			IOIn/out	CONNECTOR	IOA	FNXT	FOUT	FTsk		DMux	E		
XEROX PARC		Project Dorado	Reference DispM Board Layout					File DispM26.sil	Designer K. Pier		Rev Cg	Date 7/20/81	Page 26					



DispY	SIZE	DispM
Altem	8	Altem 85-88-89-92-93-96-97-100
Bltem	8	Bltem 101-104-105-108-109-112-113-116
CursorData	1	CursorData
AltemClkEn	1	AltemClkEn
BltemClkEn	1	BltemClkEn
AOff	1	AOff
BOff	1	BOff
HSync	1	HSync
HBlank	1	HBlank
HalfLine	1	HalfLine
VSynC	1	VSynC
VBlank	1	VBlank
RawPixelClk	1	RawPixelClk
Crystal	1	Crystal
PixelClkVCO	1	PixelClkVCO
Modes	3	Modes 28-32-33
XHSync	1	XHSync
XVSync	1	XVSync
XSyncEn	1	XSyncEn
ABypass	1	RefIn



Plus seven wire interface cable. See page 11.

DDC Slow IO System

DEVICE	TIOA	I/O	TASK	FORMAT and COMMENTS
AStatics	367	O	AHT	
NLCB	366	O	AHT	
BMap	365	O	CHT, EMU	Keep,Write,LoadAddr,0,DataOrAddr8-15
AHTFlag	364	O	AHT	
AWT	363	O	AWT	
CMap	362	O	CHT, EMU	Keep,Write,LoadAddr,0,DataOrAddr8-15
MIXER	361	O	CHT, EMU	Keep,Write,LoadAddr,0,Data.4-15
PIXELCLK	360	O	CHT, EMU	Pixel clock rate
STATUS	361	I	CHT, EMU	MType.0-3,Green.0-7,Red.0-3
STATUS	360	I	CHT, EMU	Keyboard,1,1,1,Red.4-7,Blue.0-7

00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	
				Pixel Clock Rate								Clock Divider				PIXELCLK
Keep Mixer'	Write Mixer'	Load Mixer Addr	X		Addr.0	Addr.1	Addr.2	Addr.3	Addr.4	Addr.5	Addr.6	Addr.7	Addr.8	Addr.9	Hi/Lo select	MIXER
				Mixer Data 12 bits												
Keep BMap'	Write BMap'	Load Bmap Addr	X					Address.0-7 OR Data.0-7								BMap
Keep CMap'	Write CMap'	Load CMap Addr	X					Address.0-7 OR Data.0-7								CMap
														AWT Shut Up	AHT Shut Up	AStatics
NLCB Addr 00	NLCB Addr 01	NLCB Addr 02	NLCB Addr 03	NLCB DATA 12 Bits												NLCB
											IOFetch				Set/Clr Cur WCB Flag	AWT
														X	Must Be 0	AHTFlag

1	jumper wire	16
2	0.1mFarad	15
3	3 KOhm	14
4		13
5	180 Ohm	12
6	3 KOhm	11
7		10
8		09

locations

g18,g21,j21

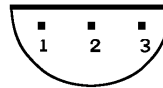
3 identical copies

1	0.1mFarad	16
2	in 3 78L05	15
3	gnd 2	14
4	out 1	13
5	0.1mFarad	12
6	12 mHnry	11
7	2 ohm	10
8	+ 22 mF	09

locations

j23,g23,g20

3 identical copies



1. Output
2. Ground
3. Input

Bottom view of pinout
for 78L05

BOTTOM VIEW!!
BOTTOM VIEW!!

SIP in location g41 is 100 ohm terminator with leg 6 cut (making DDMTIOA = 360B)

SIP in location b52 is 100 ohm terminator with legs 3 and 4 cut for Task 9D = 11B

SIPs in locations d42 and e42 are 220 ohm value instead of 100 ohm (terminators for 7 wire interface lines)

Crystal oscillators, type K1115A:
location c5, value 10 MHz, for VCO
location d13, value 20 Mhz for Alto
value 50 MHz for LF

Horizontal PROM, type MC149

location a16 and b20 for LF display
location b20 ONLY for Alto style display
programmed for each monitor type

1	8.2 KOhm	16
2	Bead	15
3	4700 pF	14
4	22 microF +	13
5	8.2 KOhm	12
6	+ 22 microF	11
7	4700 pF	10
8	Bead	09

location a1

1		16
2	2.2 KOhm	15
3	2.2 KOhm	14
4	2200pF	13
5	2200pF	12
6	2.2 KOhm	11
7	2.2 KOhm	10
8		09

location a2

1		16
2	4.7 KOhm	15
3	+ MV1403-	14
4	+ MV1403-	13
5	180 nH	12
6	15 pF	11
7	1 KOhm	10
8		09

location a4

1	jumper wire	16
2	0.1microF	15
3	0.1microF	14
4	+ 22 microF	13
5	1 KOhm	12
6	Bead	11
7		10
8		09

location a6

1		16
2	jumper wire	15
3		14
4		13
5		12
6		11
7		10
8		09

location c3